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(54) **APPARATUSES AND METHODS FOR
IMPLANTING GLENOID PROSTHESES**

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(57)

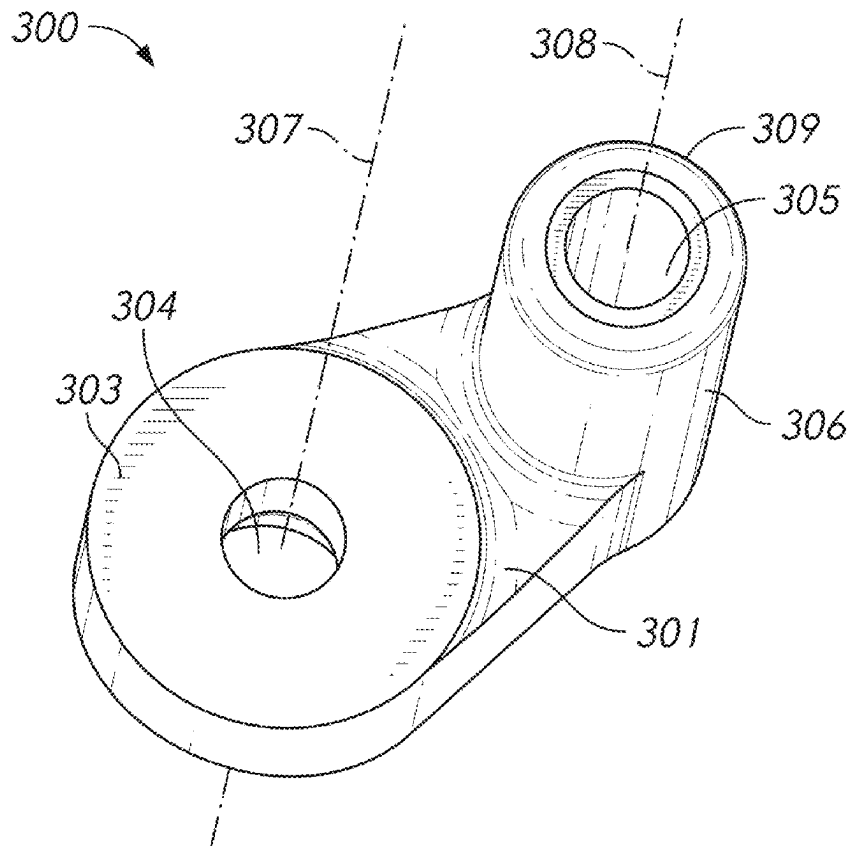
ABSTRACT

(22) Filed: **May 21, 2025**

Related U.S. Application Data

(60) Division of application No. 18/454,203, filed on Aug. 23, 2023, now Pat. No. 12,329,659, which is a division of application No. 17/172,789, filed on Feb. 10, 2021, now Pat. No. 11,779,471, which is a continuation-in-part of application No. PCT/US2020/045211, filed on Aug. 6, 2020.

Various apparatus and methods for implanting glenoid prostheses are disclosed. In various embodiments, a kit for shoulder surgery can include a partial reaming guide having a patient-matched surface shaped to conform to a scapula of a patient, the partial reaming guide configured to limit glenoid reaming about a reaming axis to only a portion of a glenoid; and an anchor peg channel guide having a patient-matched surface shaped to conform to a portion of the glenoid, the anchor peg channel guide having a channel offset from the reaming axis.



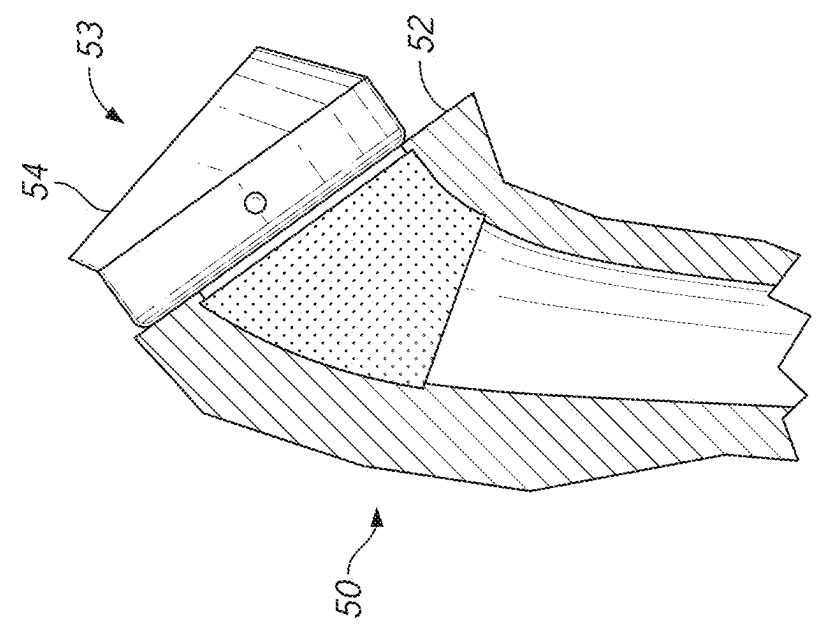
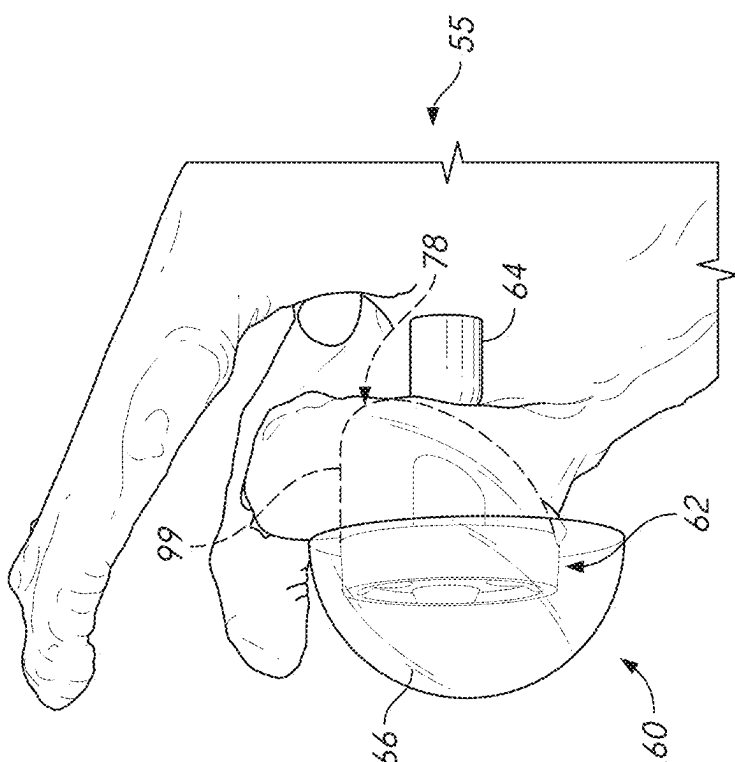


FIG. 1

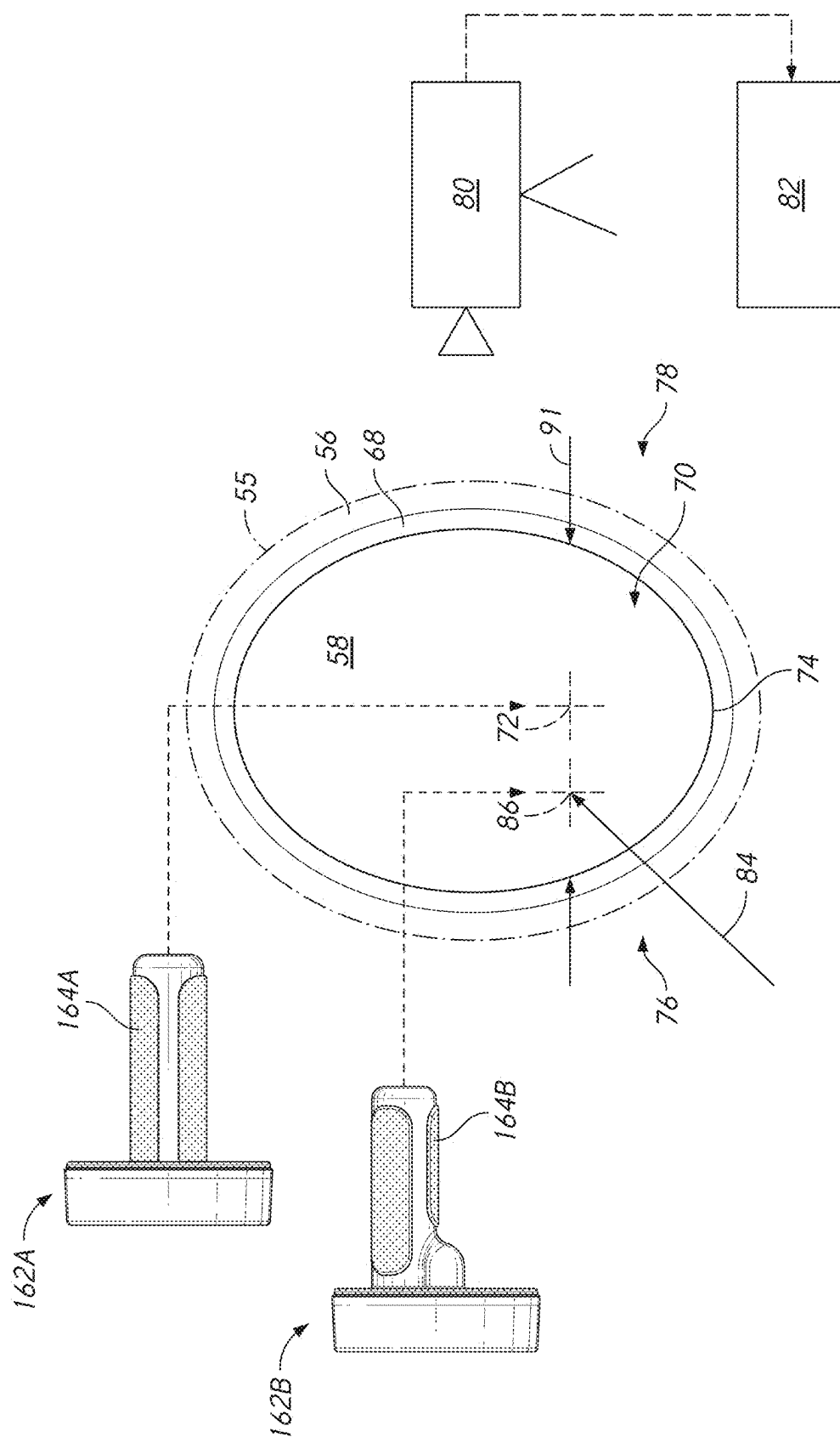


FIG. 1A

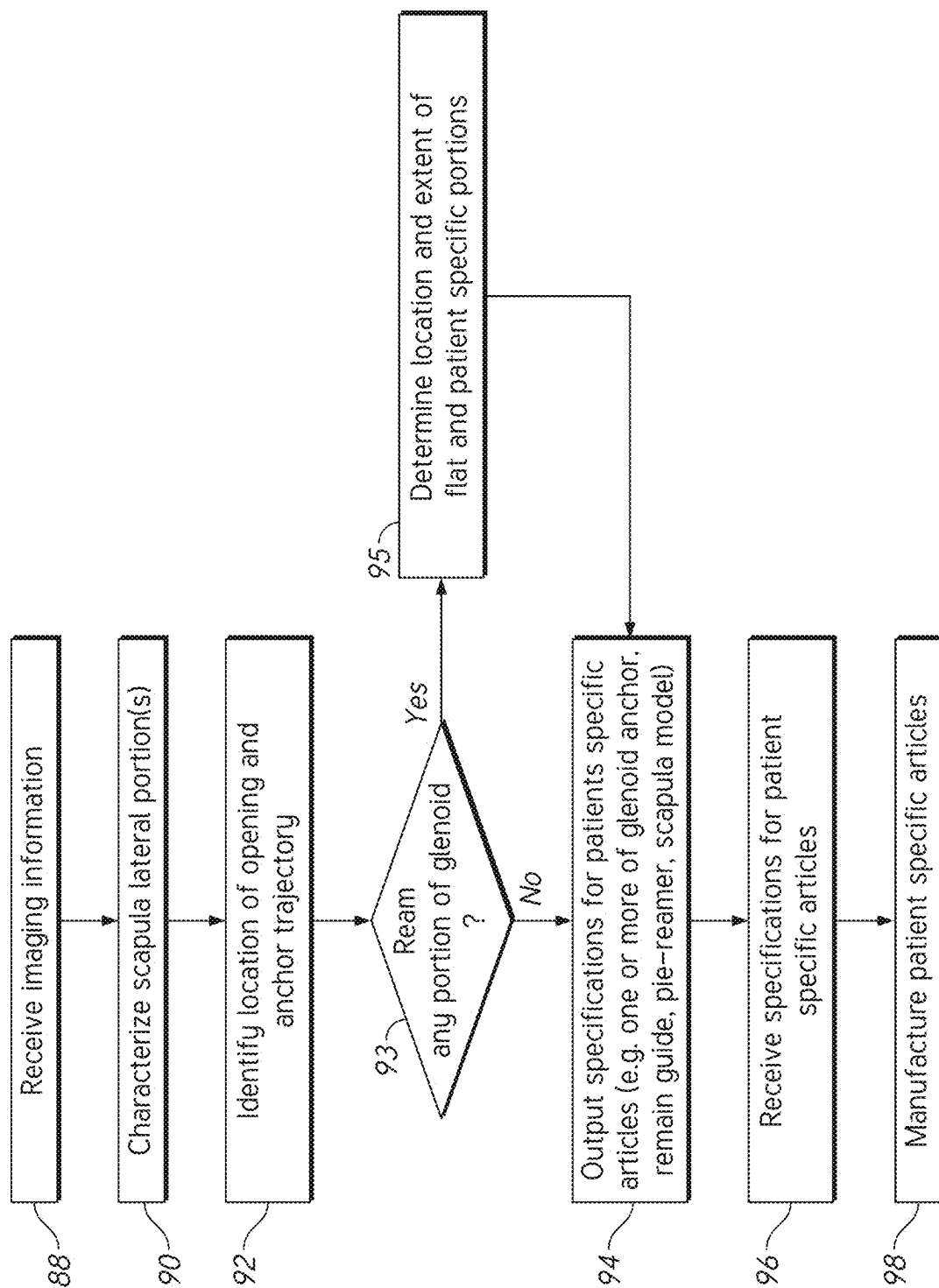


FIG. 1B

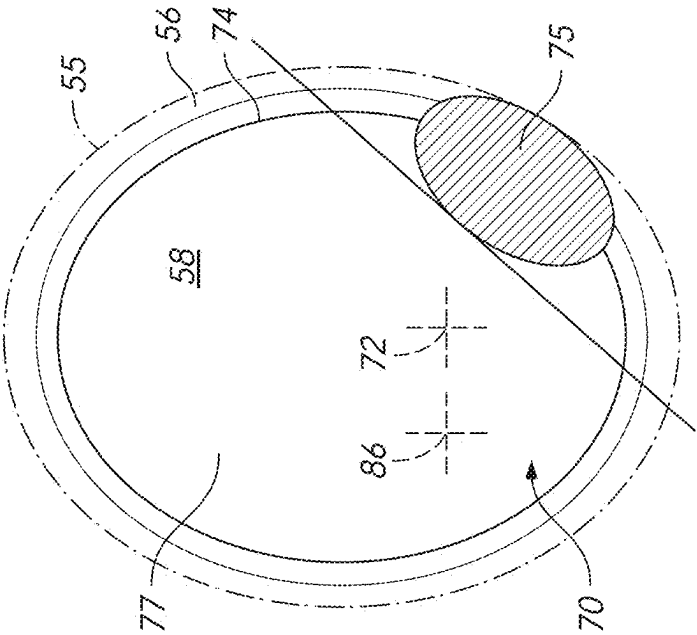


FIG. 1C

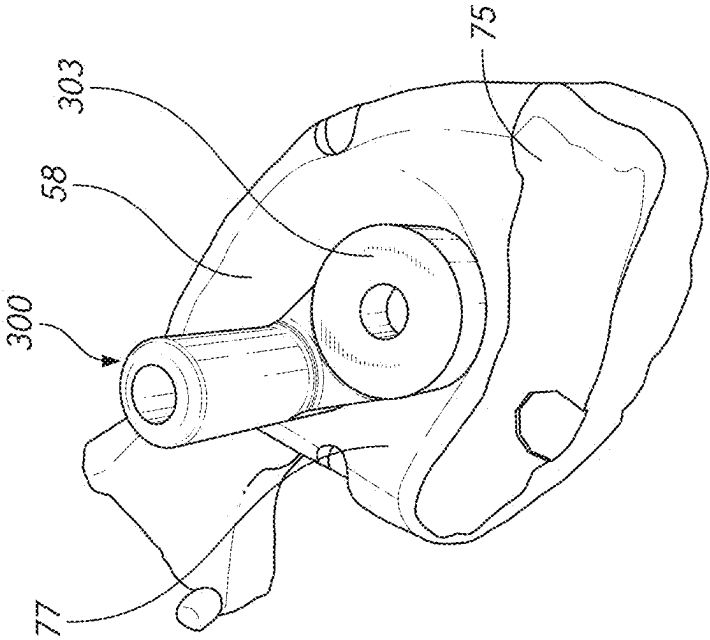


FIG. 3

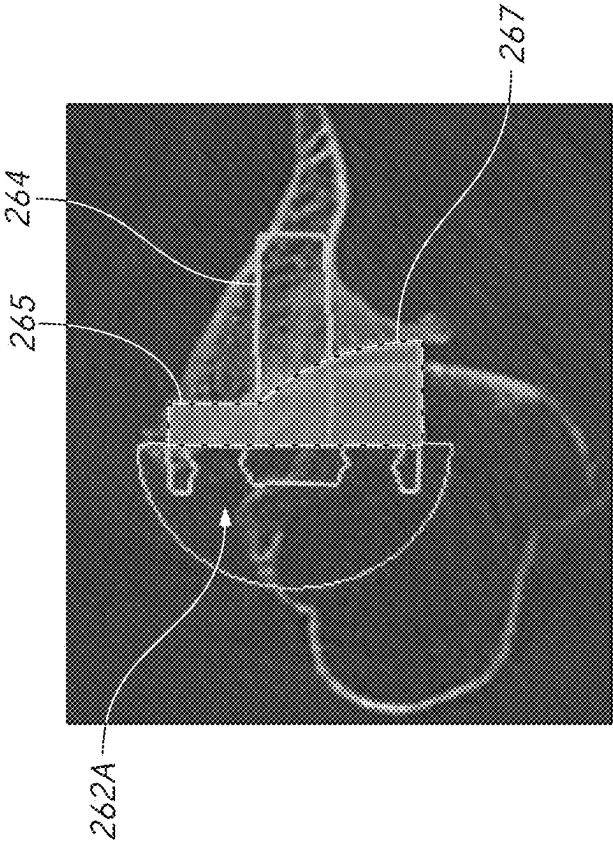


FIG. 2

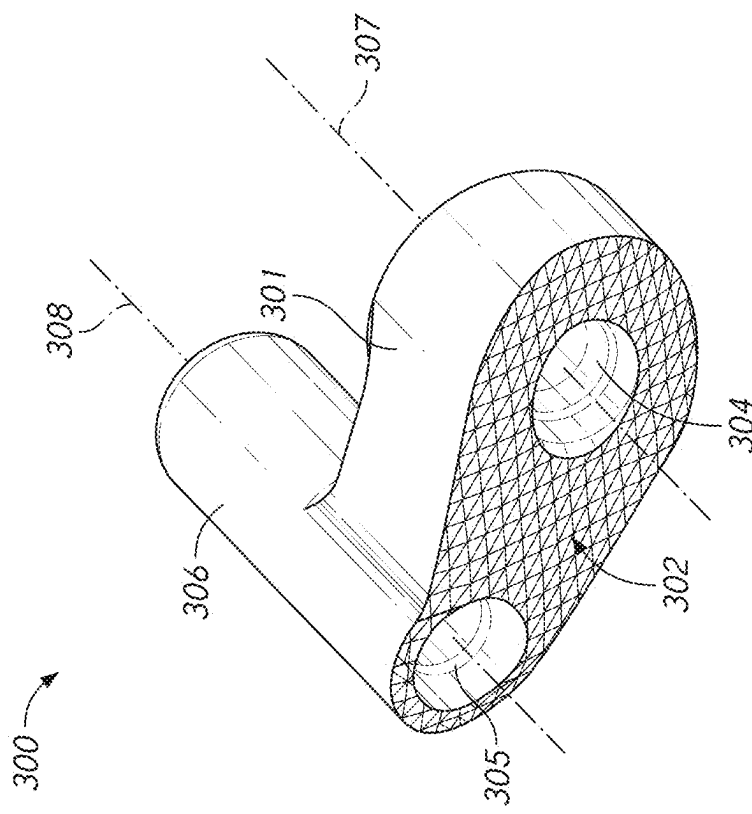


FIG. 3B

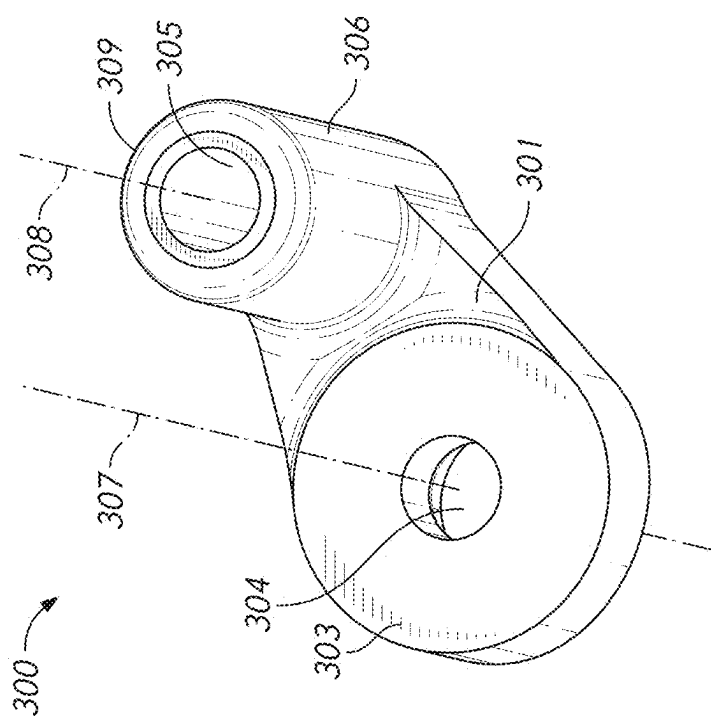


FIG. 3A

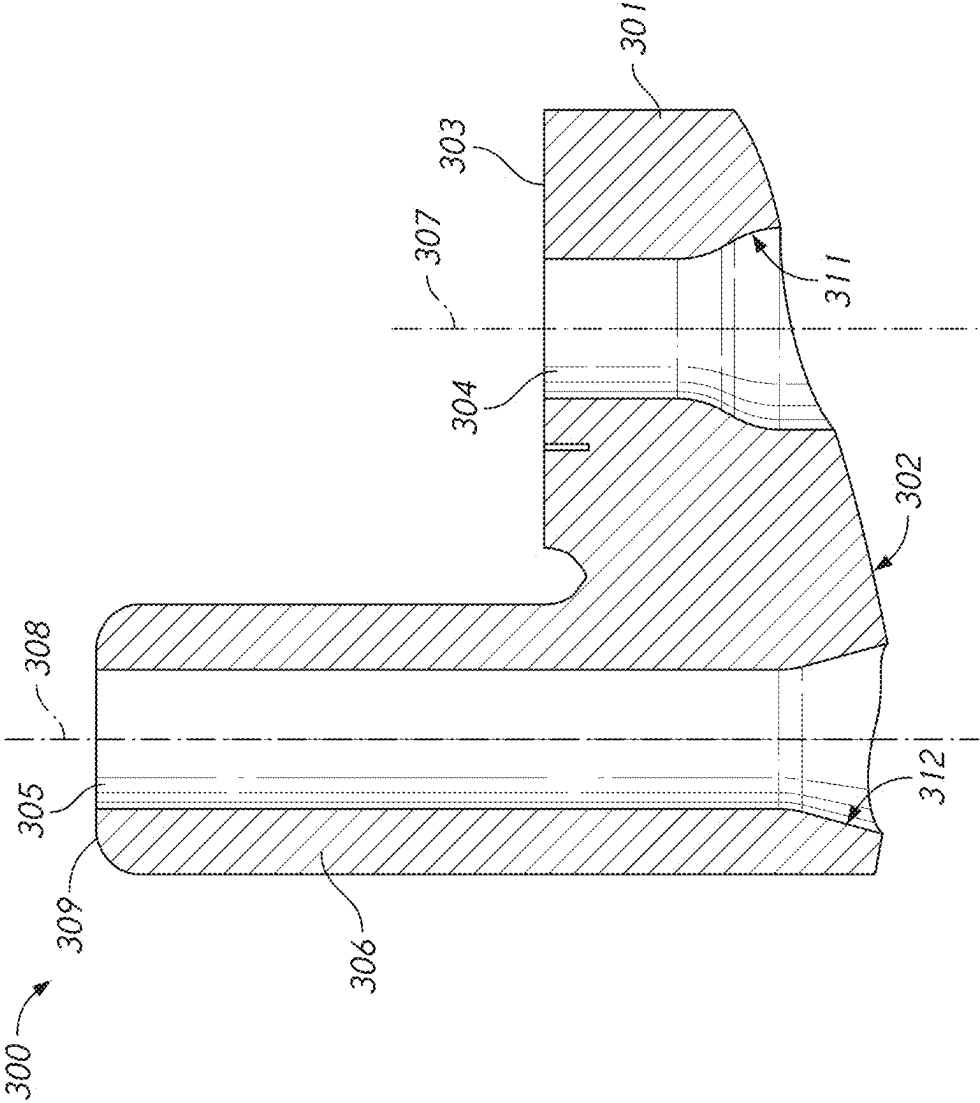


FIG. 3C

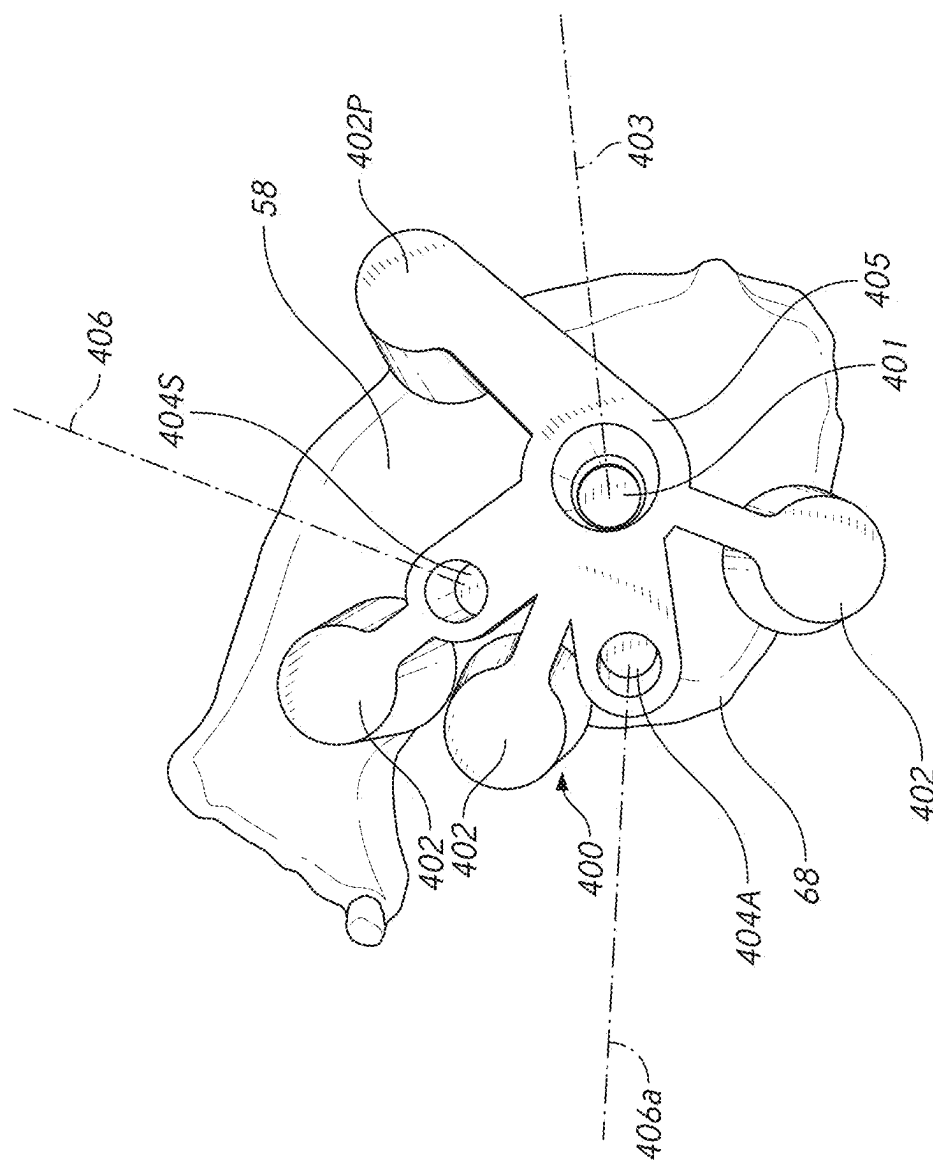


FIG. 4

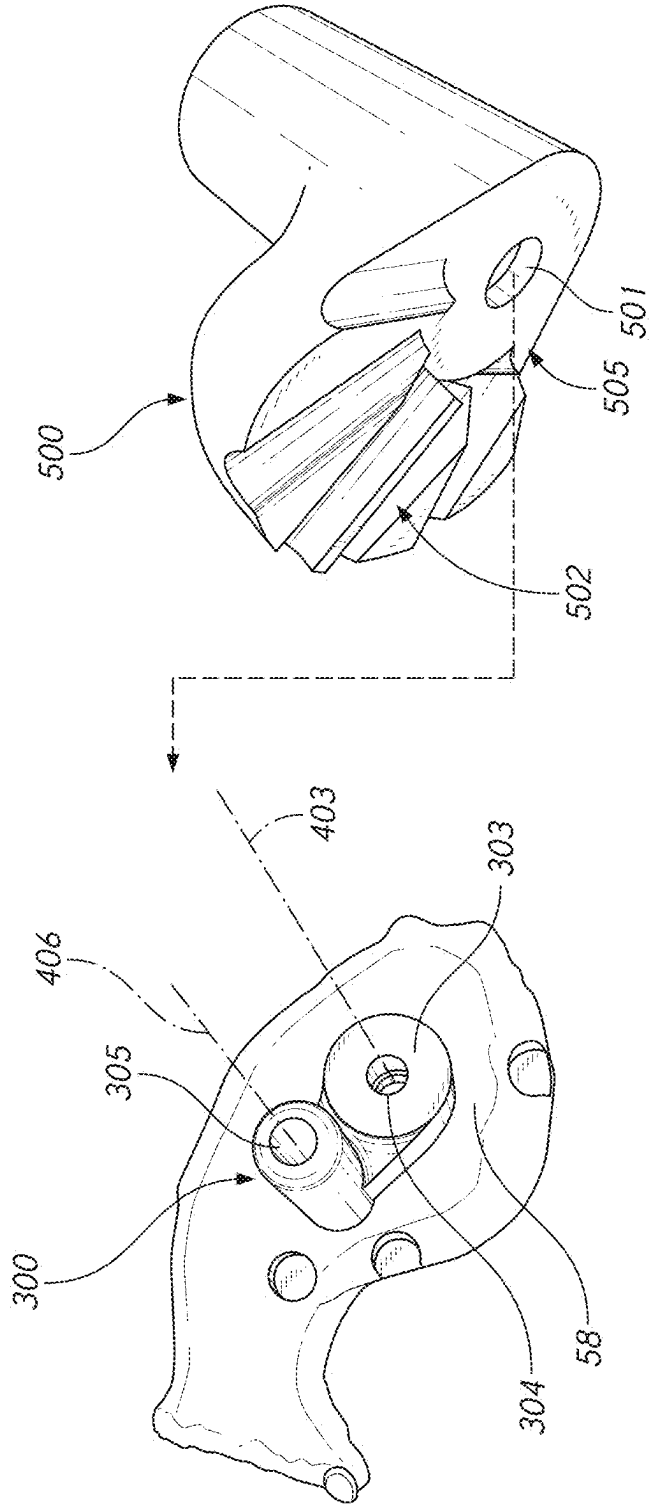


FIG. 5

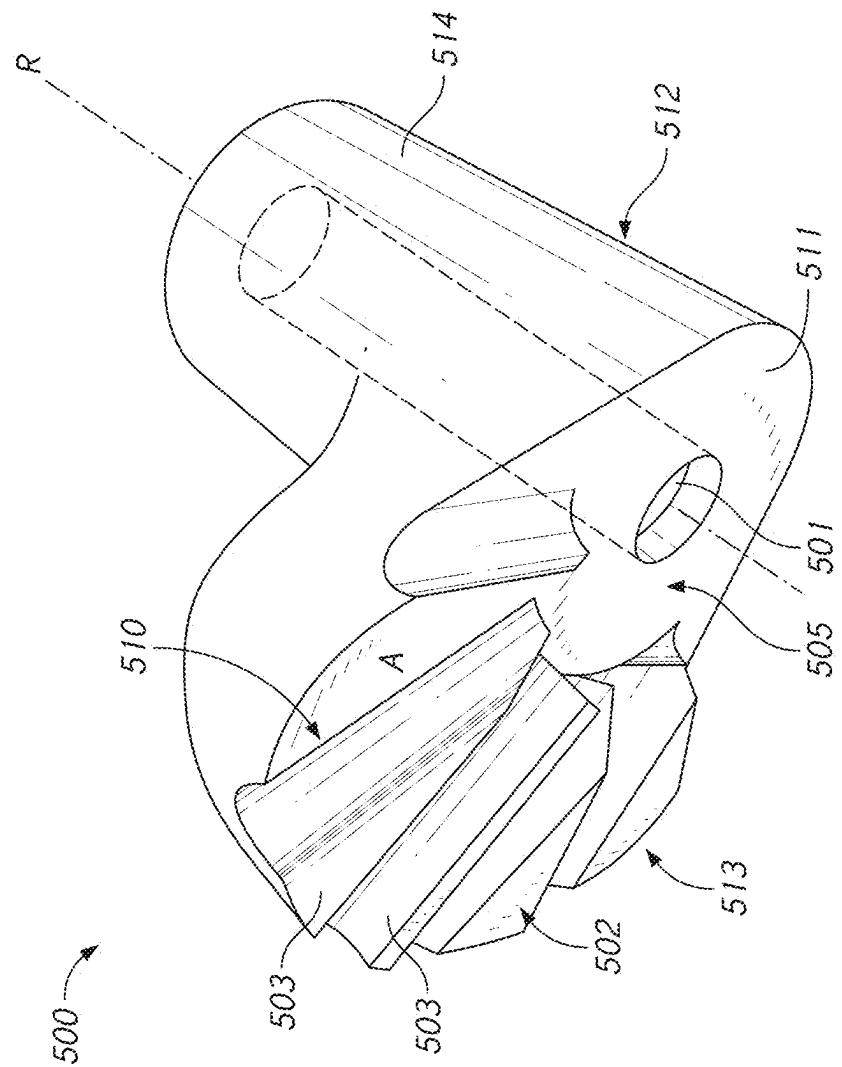


FIG. 5A

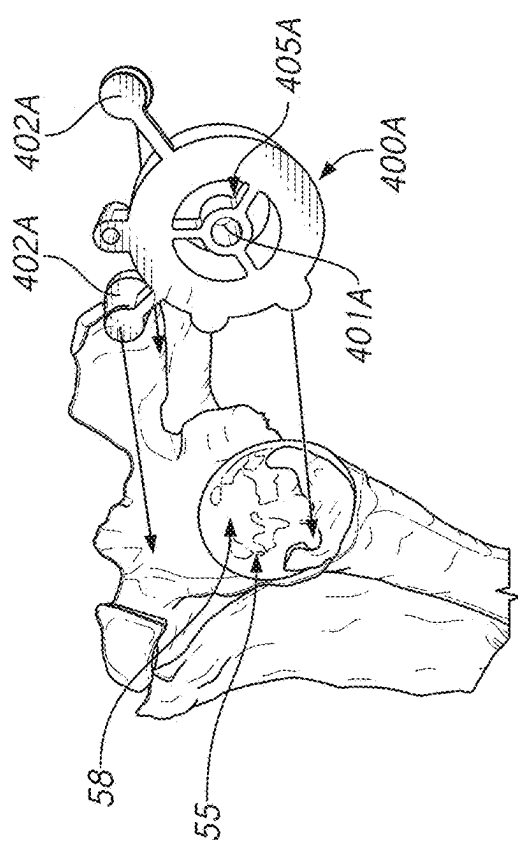


FIG. 5B

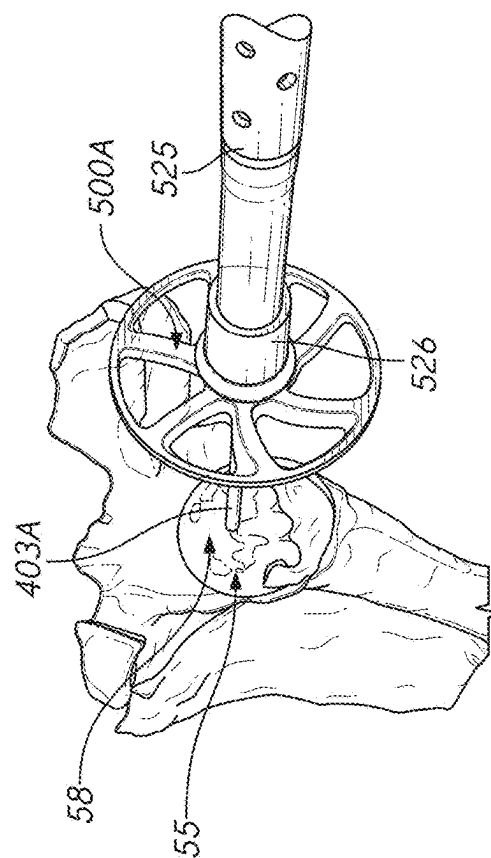


FIG. 5C

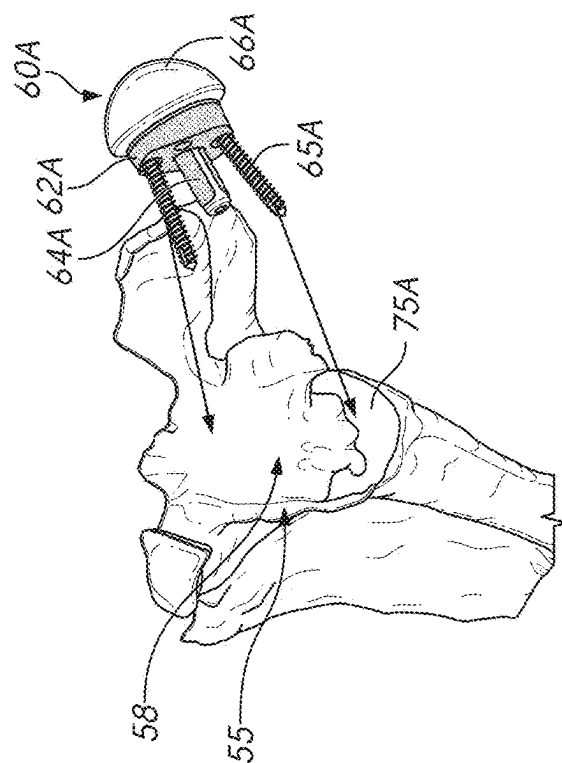


FIG. 5D

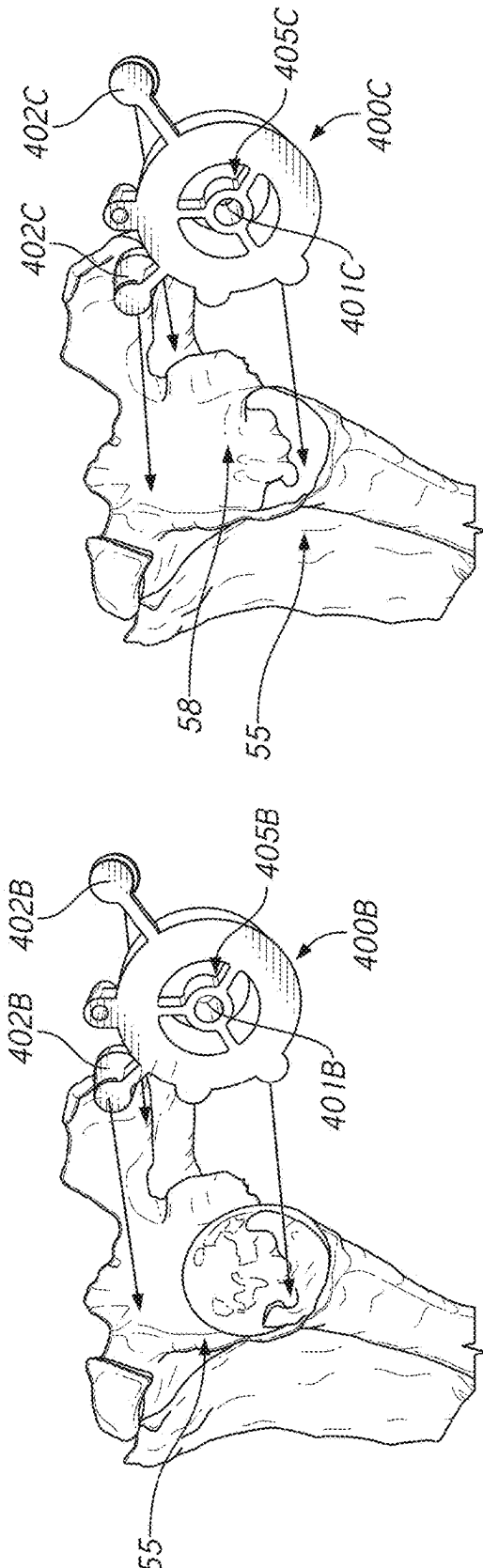


FIG. 5G

FIG. 5E

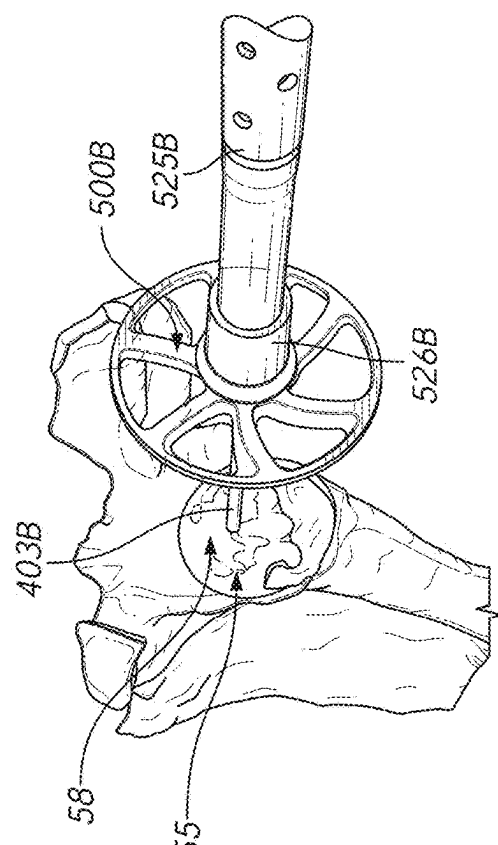


FIG. 5F

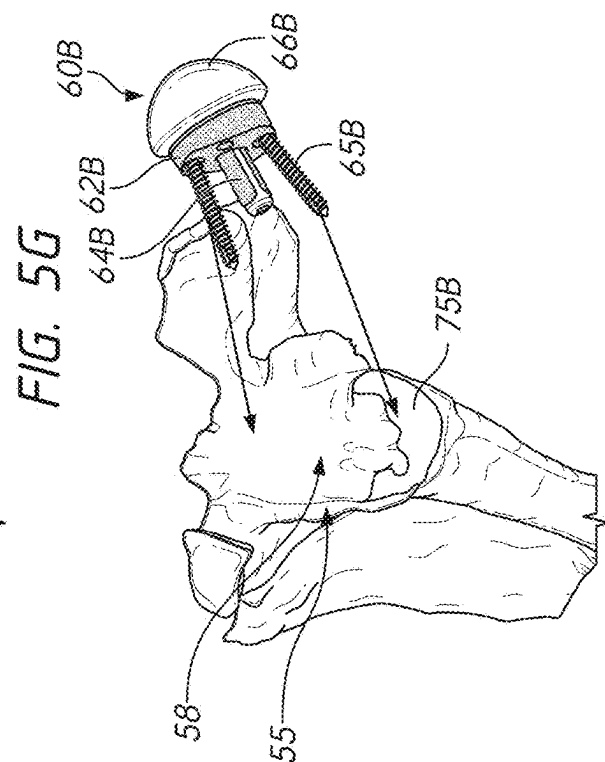


FIG. 5H

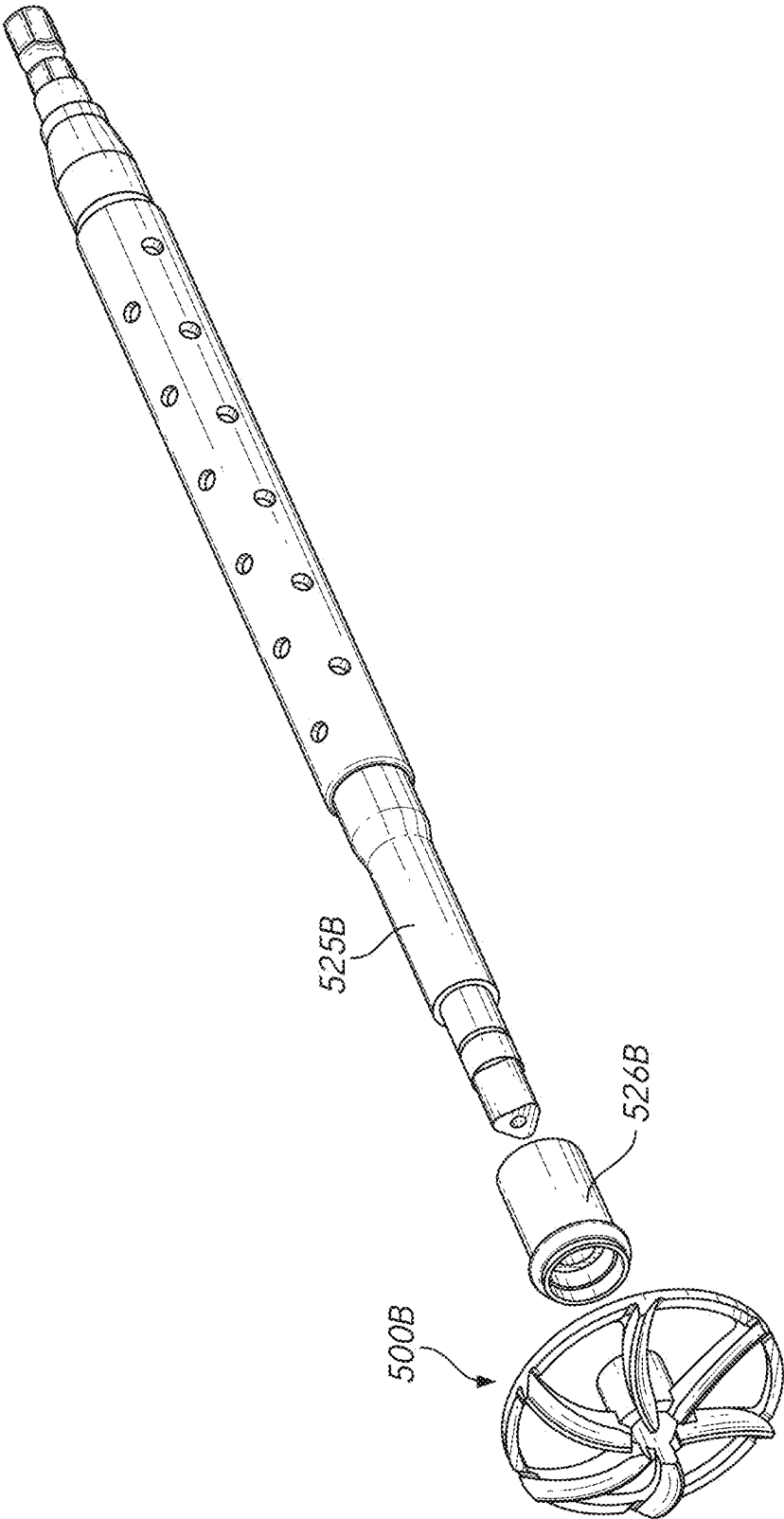


FIG. 5I

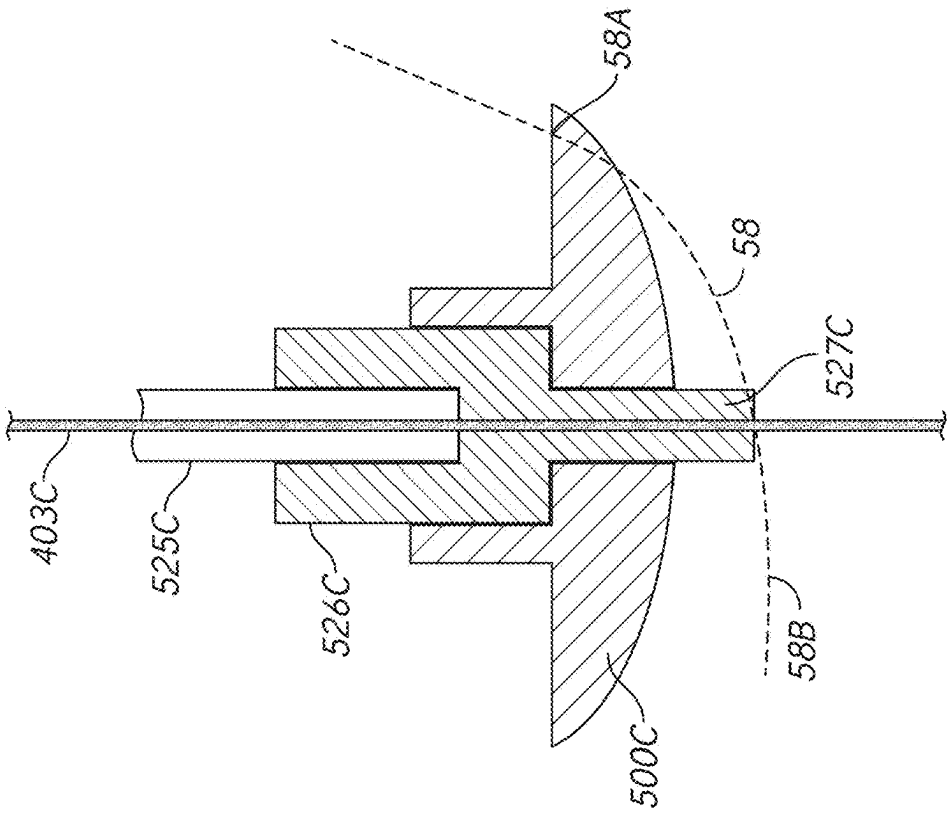


FIG. 5J

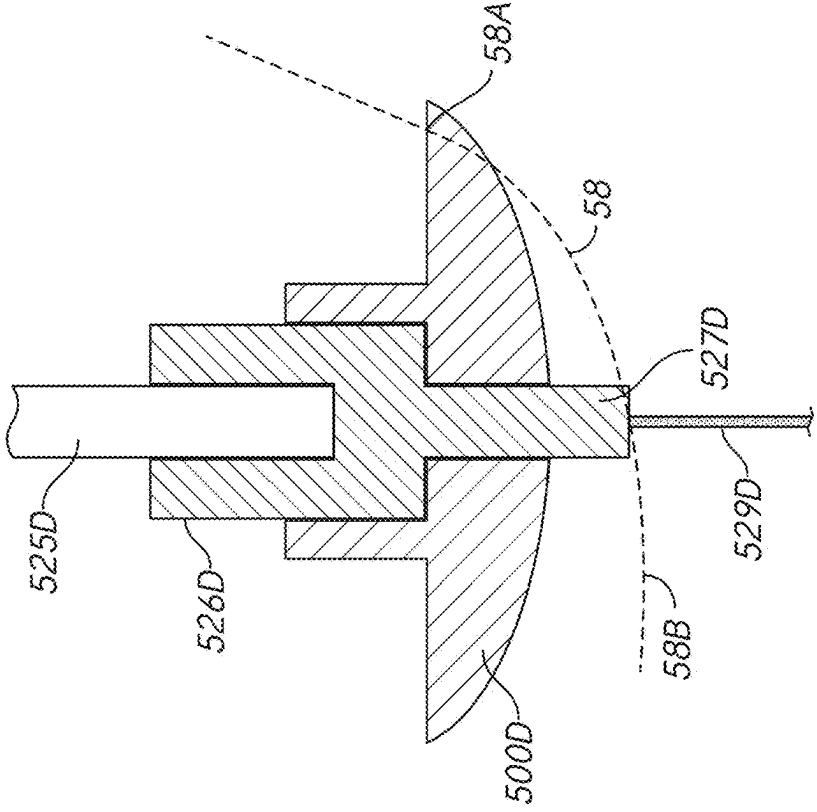


FIG. 5K

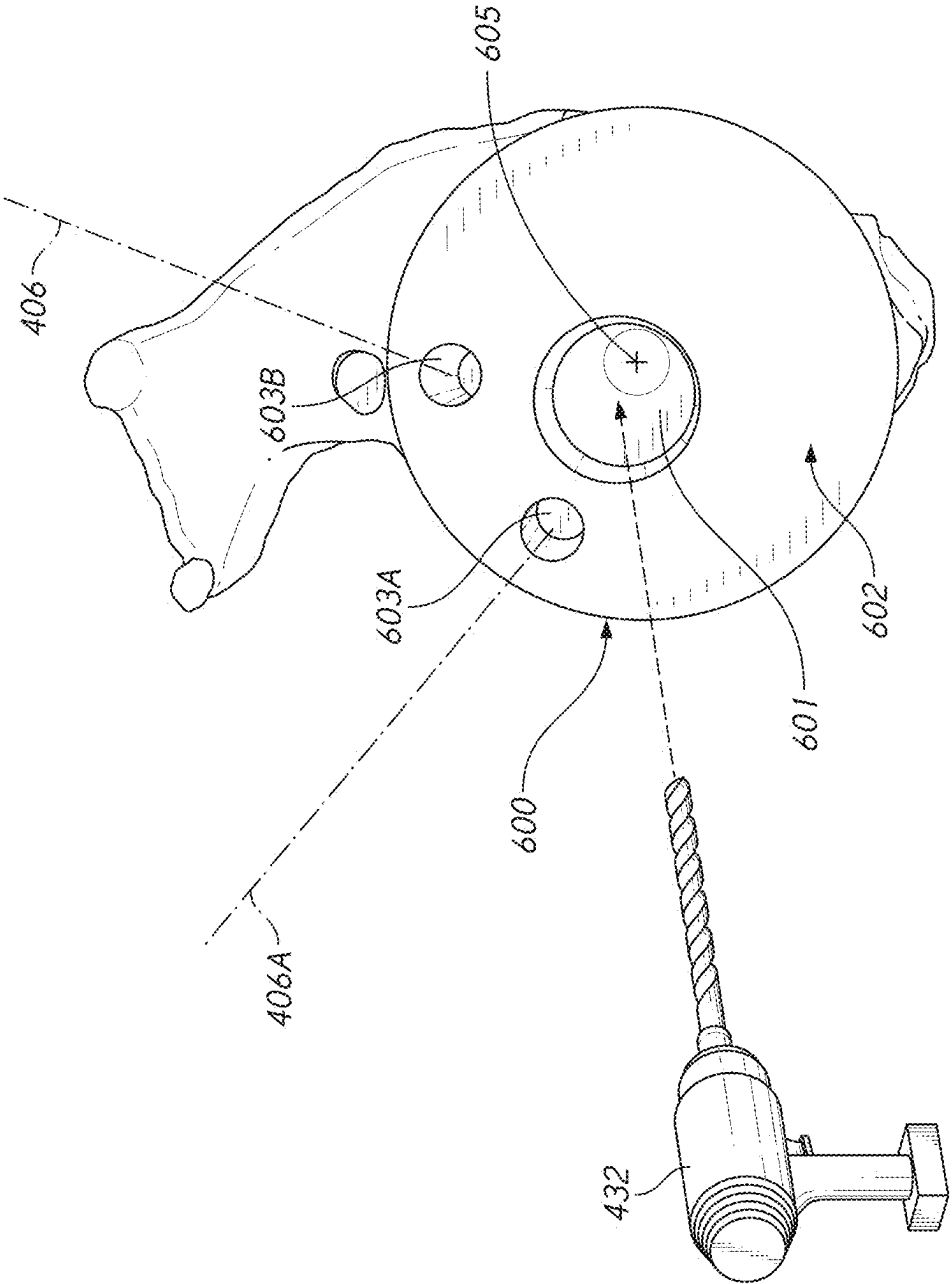


FIG. 6

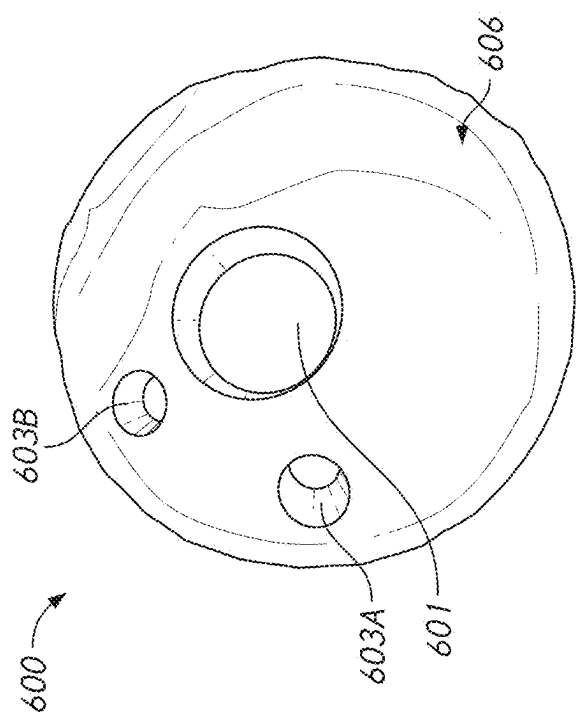


FIG. 6A

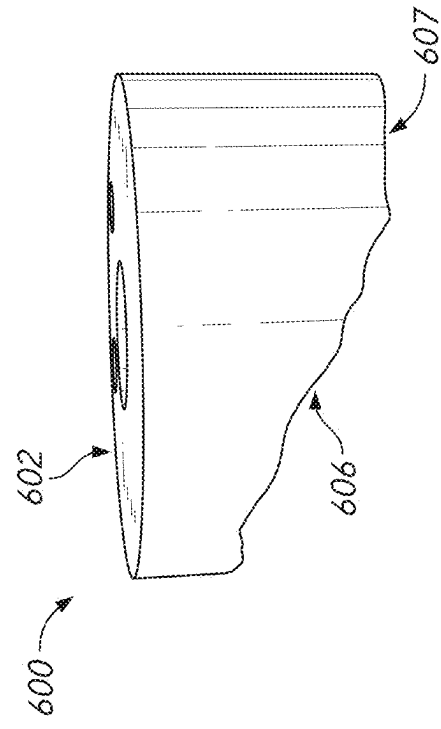


FIG. 6B

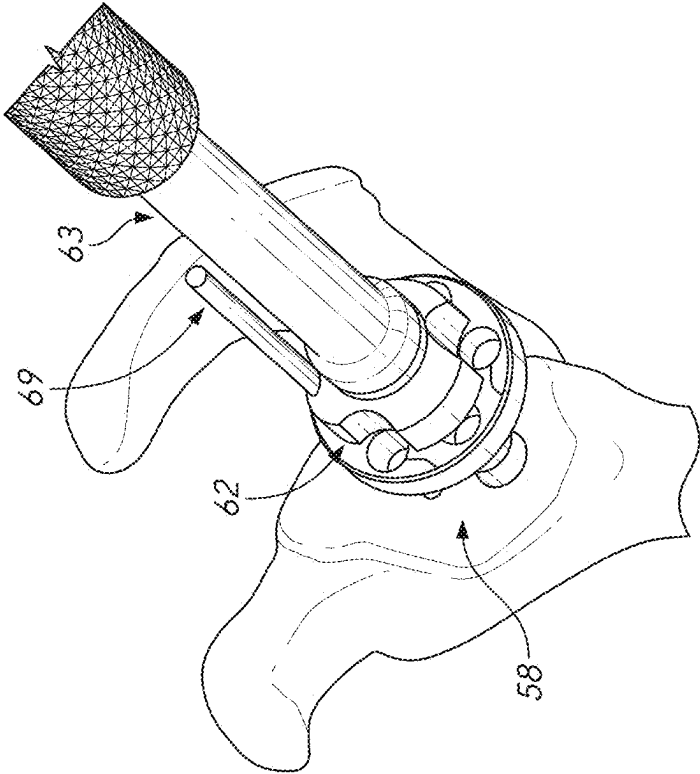


FIG. 7

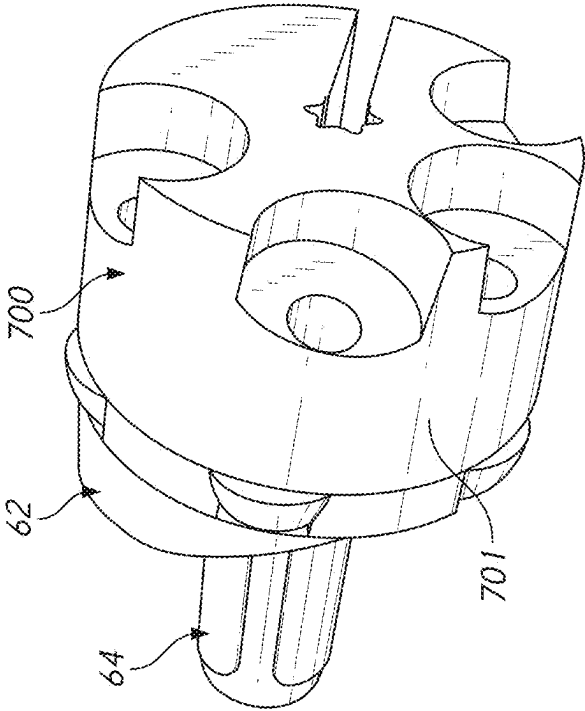


FIG. 8

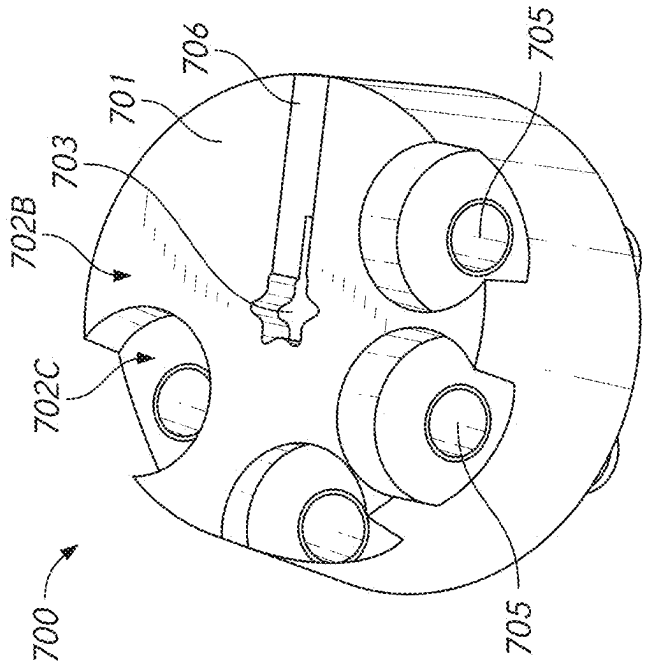


FIG. 8B

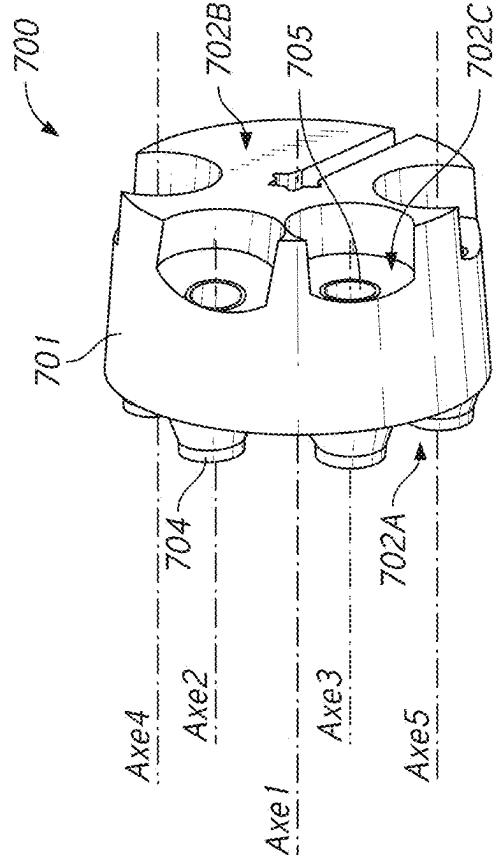


FIG. 8A

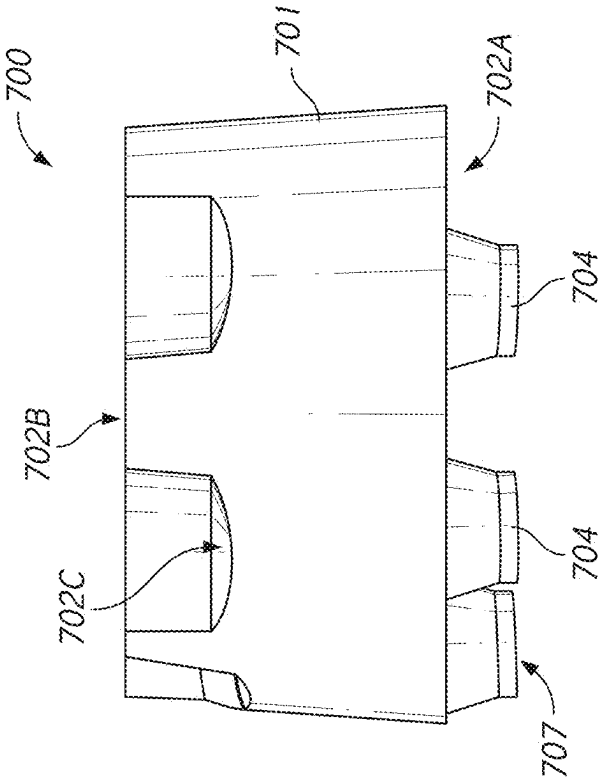


FIG. 8C

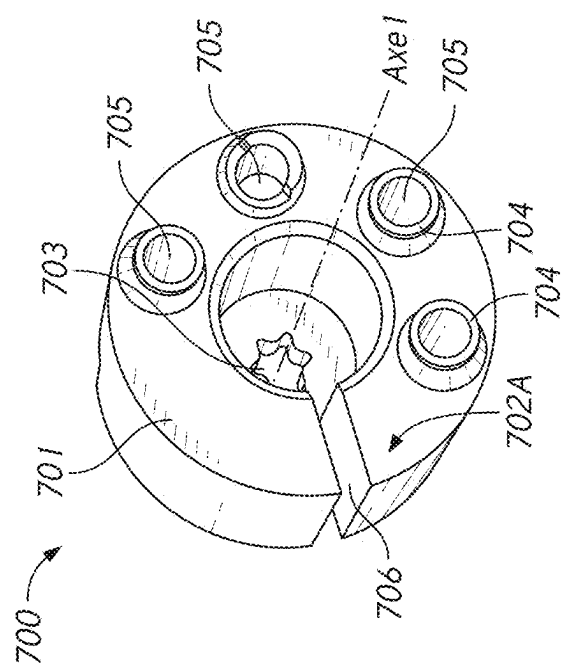


FIG. 8D

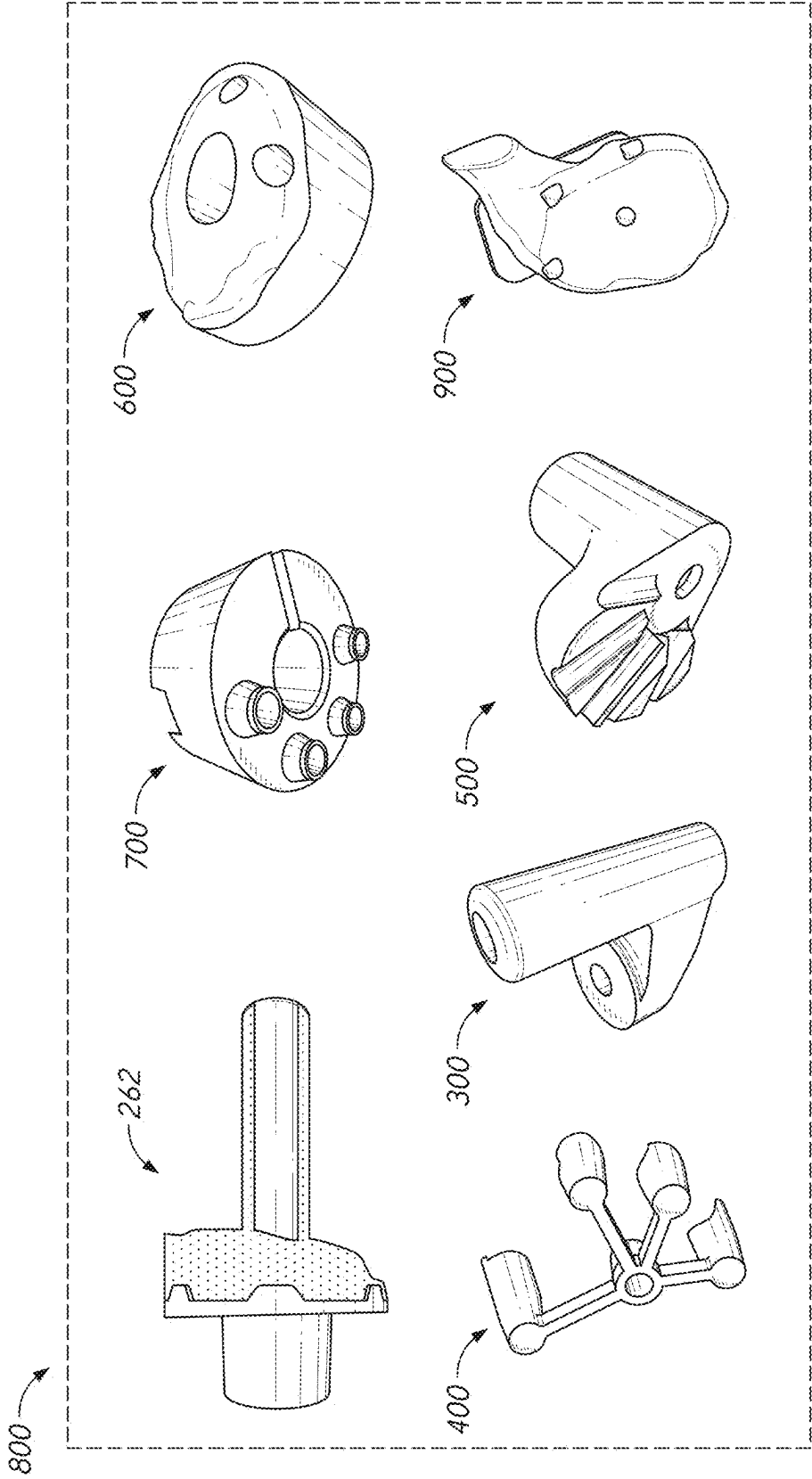


FIG. 9

APPARATUSES AND METHODS FOR IMPLANTING GLENOID PROSTHESES

INCORPORATION BY REFERENCE TO ANY PRIORITY APPLICATIONS

[0001] This application is a divisional application of co-pending U.S. patent application Ser. No. 18/454,203, filed Aug. 23, 2023, which is a divisional application of co-pending U.S. patent application Ser. No. 17/172,789, filed on Feb. 10, 2021 (now U.S. Pat. No. 11,779,471), which is a continuation-in-part of International Application No. PCT/US2020/045211, filed on Aug. 6, 2020, which claims benefit under 35 U.S.C. § 119(e) to U.S. Provisional Application No. 62/885,033, filed on Aug. 9, 2019. Any and all applications for which a foreign or domestic priority claim is identified in the Application Data Sheet as filed with the present application are hereby incorporated by reference under 37 C.F.R. §§ 1.57; 1.97; and 1.98 in their entireties.

BACKGROUND OF THE INVENTION

Field of the Invention

[0002] This application is directed to apparatuses and methods for improved preparation of a glenoid region of a scapula in connection with implantation of a shoulder prosthesis and to apparatuses that can be implanted following use of such apparatuses and methods.

Description of the Related Art

[0003] Shoulder joint conditions can sometimes be resolved with shoulder arthroplasty. More and more, efforts are being focused on making total shoulder joint arthroplasty available to patients who would benefit from such treatment. In a total shoulder joint arthroplasty, the glenoid is typically reamed and a glenoid articular component is mounted to the scapula following reaming. The articular component provides a smooth surface for movement of a humeral head or humeral articular component.

[0004] A glenoid baseplate can be used to support the glenoid articular component on the scapula. The glenoid baseplate can include an anchor peg on the medial side thereof that is configured to be inserted into scapular bone as part of securing the glenoid baseplate to the scapula.

SUMMARY OF THE INVENTION

[0005] Apparatuses and methods for improved glenoid preparation are needed to improve the placement of glenoid baseplates. For example, apparatuses and methods disclosed and claimed herein can improve glenoid preparation by reducing the amount of bone removed prior to implanting a glenoid baseplate. Also, apparatuses and methods disclosed and claimed herein can improve glenoid preparation by providing flexibility in the location of an anchor peg of the baseplate, e.g., at or spaced from a central position of the glenoid of a particular patient. Apparatuses and methods disclosed and claimed herein can improve glenoid preparation by allowing a particular patient to benefit from reduced, minimal or no reaming in once region of a glenoid and to allow a surgeon to ream another region of the glenoid such as to remove obstructive osteophytes or other problematic bone formations. Additional improvements over the prior art are described and claimed herein below.

[0006] In one embodiment, a method for performing shoulder surgery is disclosed. The method can include guiding a guide pin into the glenoid surface along a reaming axis. The method can include placing a partial reaming guide in contact with the glenoid surface over the guide pin. A reamer can be advanced over the guide pin to ream the glenoid surface. The reamer can be further advanced over the guide pin until the reamer contacts the partial reaming guide, whereby such contact limits reaming to only a portion of the glenoid surface.

[0007] In some embodiments, the method can include forming a channel in the scapula medially from the glenoid surface, the channel configured to receive an anchor peg of a glenoid baseplate. A drill can be advanced over the guide pin to form an anchor peg channel centered on the reaming axis. The method can include advancing an anchor peg channel forming guide toward the glenoid surface, the anchor peg channel forming guide comprising a body and an aperture formed inward of a periphery of the body and securing the anchor peg channel forming guide against the glenoid surface with the aperture off-set from the reaming axis. In some embodiments, advancing the reamer and further advancing the reamer comprises reciprocating a reaming surface of the reamer about an angle of less than 180 degrees relative to the reamer axis. In various embodiments, the method can further include inserting an anchor peg of a glenoid baseplate into an anchor peg channel formed in the glenoid surface. A screw trajectory guide can be coupled with the baseplate and one or more screw holes can be formed in the scapula through the screw trajectory guide and the baseplate. A depth of the one or more screw holes can be controlled with a corresponding depth control surface of the screw trajectory guide. The method can further include defining a reaming axis based on image data responsive to a scan of a scapula of a patient.

[0008] In another embodiment, a kit for shoulder surgery is disclosed. The kit can include a partial reaming guide having a patient-matched surface shaped to conform to a scapula of a patient. The partial reaming guide can be configured to limit glenoid reaming about a reaming axis to only a portion of a glenoid. The kit can include an anchor peg channel guide having a patient-matched surface shaped to conform to a portion of the glenoid. The anchor peg channel guide can have a channel offset from the reaming axis.

[0009] In some embodiments, the kit can include a screw trajectory guide having a protrusion shaped to be inserted into an aperture of a glenoid baseplate, the protrusion including a channel extending therethrough. The kit can include a glenoid baseplate having an anchor member. The kit can include a three-dimensional (3D) model of the scapula of the patient. The kit can include an alignment guide having a plurality of contact members configured to conform to a plurality of surfaces of a scapula, the alignment guide comprising an aperture configured for placing a guide wire in the glenoid. The anchor peg channel guide can have a portion configured to rest on the reamed portion of the glenoid. The kit can include a reaming device comprising a stop surface and a reaming portion having one or more reaming features configured to ream a bone surface. The reaming device further can include a guide hole through the reamer body, the guide hole sized and shaped to receive a guide pin therethrough. The reaming portion can include an arc-shaped structure that delimits an angle less than 180

degrees. The reaming device can include a first lower portion which comprises the reaming portion and a second upper portion angled relative to the lower portion, the guide hole formed through the second upper portion.

[0010] In another embodiment, a partial reaming guide for use in a shoulder treatment procedure is disclosed. The partial reaming guide can include a guide body comprising a patient-matched surface shaped to conform to a portion of a scapula of a patient. The partial reaming guide can include a reamer depth stop surface at a first height above the patient-matched surface, the reamer depth stop surface positioned to serve as a depth stop for the reaming device to control a depth of reaming. The partial reaming guide can include a first hole through the guide body extending from the reamer depth stop surface to the mounting surface, the first hole aligned with a reaming axis.

[0011] In some embodiments, the guide body can include a raised surface at a second height greater than the first height, a second hole extending through the guide body from the raised surface to the patient-matched surface. The guide body can include a receiver body defined between the raised surface and the mounting surface. A second hole can be formed through the guide body offset from the first hole, the second hole to rotationally orient the guide body relative to the scapula.

[0012] In another embodiment, an anchor peg channel guide for use in a shoulder treatment procedure is disclosed. The anchor peg channel guide can include a guide body comprising a patient-matched surface shaped to conform to a scapula of a patient and a lateral surface opposite the mounting surface. The anchor peg channel guide can include a channel disposed through the guide body, the channel positioned to be offset from a reaming axis of the scapula. The guide body can include a rotational alignment hole through the guide body.

[0013] In some embodiments, a center of a periphery of the guide body of the anchor peg channel guide can be spaced apart from a center of the channel. At least one peripheral hole can be provided for securing the guide body to the scapula.

[0014] In another embodiment, a screw trajectory guide for use in a shoulder procedure is disclosed. The screw trajectory guide can include a guide body having a first surface shaped to mate with a glenoid baseplate, a second surface opposite the first surface, and a third surface recessed from the second surface between the first and second surfaces. The guide body can include a protrusion extending from the first surface and shaped to be inserted into corresponding apertures of the glenoid baseplate. The guide body can include a channel extending from the third surface through the protrusion to a distal end of the protrusion.

[0015] In some embodiments, a guide channel can be formed through the guide body, the guide channel to receive a guide wire therethrough. A slot can extend from the guide channel to an outer periphery of the guide body. A plurality of protrusions can extend from the first surface and be shaped to be inserted into corresponding apertures of the glenoid baseplate. A plurality of channels can also be provided, with each channel extending from one of a plurality of third surfaces through a corresponding one of a plurality of protrusions. At least one of the third surfaces can be disposed at an elevation that is prescribed for the patient

to control a depth of a peripheral screw hole formed in a scapula through the channel extending from the at least one third surface.

[0016] In another embodiment, a reamer for use in shoulder surgery is disclosed. The reamer can include a reamer body comprising a stop surface and a reaming portion having one or more reaming features configured to ream a bone surface. The reamer can include a guide hole through the reamer body, the guide hole sized and shaped to receive a guide pin therethrough. The reaming portion can comprise an arc-shaped structure that delimits an angle less than 180 degrees.

[0017] In some embodiments, the reamer body can include a first lower portion through which comprises the reaming portion and a second upper portion angled relative to the lower portion, the guide hole formed through the second upper portion.

[0018] In one embodiment, a method for performing shoulder surgery is disclosed. The method can include guiding a guide pin into the glenoid surface along a reaming axis; selecting a tip connector based on patient-specific image data; connecting a reaming device to a handle by way of the selected tip connector; and advancing the reaming device over the guide pin to at least partially ream the glenoid surface.

[0019] In some embodiments, the method can include after the advancing, placing a guide pin guide over the at least partially reamed glenoid surface to verify the at least partial reaming, the guide pin guide having a patient-matched bone-facing surface. In some embodiments, the method can include installing a glenoid assembly in the glenoid surface after the advancing. In some embodiments, selecting the tip connector comprises selecting the tip connector from a plurality of differently-sized tip connectors. In some embodiments, selecting the tip connector comprises forming the tip connector based on the patient-specific image data.

[0020] In another embodiment, a reaming instrument can include a reamer shaft; a reaming device; and a tip connector that connects the reaming device to the reamer shaft, the tip connector having a projection that extends distal a distal-most surface of the reaming device.

[0021] In some embodiments, the reaming instrument can include an opening through the reaming device and the tip connector, the opening sized to receive a guide pin. In some embodiments, the reaming instrument can include a plurality of tip connectors with distally-extending projections having different lengths. In some embodiments, the tip connector comprises a second projection extending distally from the projection, a diameter of the second projection smaller than a diameter of the projection. In some embodiments, a kit can comprise the reaming instrument, a first guide pin guide having a surface that is matched to a patient's native bone structure, and a second guide pin guide having a surface that is matched to a reamed portion of the patient's glenoid. In some embodiments, the kit can include a glenoid assembly including a glenoid baseplate and an articular body supported by the glenoid baseplate.

BRIEF DESCRIPTION OF THE DRAWINGS

[0022] Various embodiments are depicted in the accompanying drawings for illustrative purposes, and should in no way be interpreted as limiting the scope of the embodiments. Furthermore, various features of different disclosed embodi-

ments can be combined to form additional embodiments, which are part of this disclosure.

[0023] FIG. 1 shows a model of a human humerus and a scapula with a glenoid, the glenoid having an articular assembly coupled thereto, the articular assembly including a glenoid baseplate with a medial end that projects through a posterior wall of the scapula, the humerus having a reverse implant assembly coupled thereto;

[0024] FIG. 1A is a schematic view of a glenoid and an imager that can be used to gather imaging information pertaining to the glenoid and of two example baseplates that can be selected, designed and/or manufactured based on analysis of such imaging information;

[0025] FIG. 1B is a flow chart of a method that can be performed with imaging information;

[0026] FIG. 1C is a schematic diagram of a portion of a scapula including a glenoid, identifying a portion of the glenoid to be removed prior to implanting a baseplate;

[0027] FIG. 2 is a side perspective view of a baseplate having a portion having a patient-matched medial side and a portion configured to mate with a reamed or otherwise modified glenoid bone surface;

[0028] FIG. 3 is a schematic view of a glenoid that has been at least partially prepared for mounting a glenoid baseplate thereto, the preparation including reaming a limited area of the glenoid;

[0029] FIG. 3A is a top perspective view of one embodiment of a partial reaming guide having a reamer depth stop disposed on a lateral side thereof;

[0030] FIG. 3B is a bottom perspective view of the partial reaming guide of FIG. 3A;

[0031] FIG. 3C is a cross-section of the partial reaming guide of FIGS. 3A-3B taking along a plane perpendicular to the reamer depth stop and to longitudinal axes of two holes disposed through the guide;

[0032] FIG. 4 is a schematic view of a glenoid during a method of preparation of the glenoid, in which a guide pin guide has been placed against a rim of the glenoid and/or portions of the scapula around the glenoid and a guide pin has been placed through a guide pin hole in the guide pin guide;

[0033] FIG. 5 is a schematic view of a glenoid during a method of preparation of the glenoid, in which the partial reaming guide of FIG. 3A has been advanced over the guide pin placed in the step illustrated in FIG. 4 and further illustrating advancing a reamer of the guide pin toward the depth stop of the partial reaming guide;

[0034] FIG. 5A is a bottom perspective view of the reamer illustrated in FIG. 5;

[0035] FIGS. 5B-5D illustrate an example method of reaming a portion of a scapula, according to various embodiments;

[0036] FIGS. 5E-5H illustrate an example method of reaming a portion of a scapula, according to another embodiment;

[0037] FIGS. 5I-5K illustrate example reaming instruments, according to various embodiments;

[0038] FIG. 6 is a schematic view of a glenoid during a method of preparation of the glenoid, in which an anchor peg preparation guide has been advanced onto the glenoid, the guide being secured to the glenoid and further illustrating an instrument being advanced through the guide to form a blind hole or other opening or channel into the glenoid to receive a baseplate anchor peg;

[0039] FIG. 6A is a medial side view of the anchor peg channel forming guide illustrated in FIG. 6;

[0040] FIG. 6B is a side perspective view of the anchor peg channel forming guide illustrated in FIG. 6;

[0041] FIG. 7 is a schematic view of a glenoid during a method of implanting a glenoid baseplate into a glenoid;

[0042] FIG. 8 is a perspective view of a peripheral screw trajectory guide mated to a glenoid baseplate, which can be used to prepare peripheral screw channels;

[0043] FIG. 8A is a lateral side perspective view of the peripheral screw trajectory guide of FIG. 8 illustrating various peripheral screw trajectories;

[0044] FIG. 8B is a lateral side perspective view of the peripheral screw trajectory guide of FIG. 8 showing a slot feature for removal of the guide;

[0045] FIG. 8C is a side view of the peripheral screw trajectory guide of FIG. 8 showing depth stop surfaces and baseplate mating protrusions disposed on a medial side;

[0046] FIG. 8D is a medial side perspective view of the peripheral screw trajectory guide of FIG. 8 showing a baseplate mating channel formed therein; and

[0047] FIG. 9 illustrates one of several kits that can be provided including any two or more of the components shown therein or described herein.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENT

[0048] This application is directed to improving the success in providing sound connection between a glenoid assembly and a human scapula. These improvements are intended to allow for greater success in shoulder arthroplasty surgery. These improvements also allow a surgeon, engineer or other personnel involved in implementing a surgical treatment to optionally position a baseplate anchor peg centrally or eccentrically and also to determine whether to implement a surgery with some reaming, e.g., to remove osteophytes or other obstructive bone, while still providing a baseplate medial surface that is at least partially patient matched to reduce, minimize or eliminate reaming for such portions.

[0049] FIG. 1 illustrates concerns that can arise in some shoulder procedures, showing a humerus 50 and a scapula 55 of a shoulder having reverse shoulder implants disposed therein. The humerus 50 has a humeral resection 52. A humeral implant assembly 53 including a humeral anchor (shown beneath the humeral resection 52) and a reverse articular body 54. The reverse articular body 54 can be disposed above the humeral resection 52. The reverse articular body 54 can be at least partially below the humeral resection 52. The scapula 55 has a glenoid, which is the portion of the scapula 55 on which the head of the humerus 50 normally articulates. Following reverse shoulder arthroplasty, this function is provided by an articular body 66 that is coupled to the scapula 55. A glenoid assembly 60 can be provide that includes a glenoid baseplate 62 to support the articular body 66. The glenoid baseplate 62 can be coupled with the scapula 55. The glenoid baseplate 62 can have an anchor peg 64 configured to be advanced into the scapula 55. The glenoid baseplate 62 can be placed on the glenoid without or with minimal reaming. The glenoid baseplate 62 can be placed on the glenoid with partial reaming, using apparatuses and methods described herein.

[0050] In a sub-optimal case, the glenoid baseplate 62 is not properly placed on the scapula 55. FIG. 1 shows that the

anchor peg **64** of the glenoid baseplate **62** can be placed into the scapula **55** in a sub-optimal manner in which a medial end of the anchor peg **64** pierces the posterior surface **78** of the scapula **55**. The anchor peg **64** can be exposed outside the scapula **55** in that case. This outcome is sub-optimal for several reasons. The security of the connection between the anchor peg **64** and the bone of the scapula **55** is a function of the length over which there is direct contact between these structures. The direct contact provides opportunities for bony ingrowth, providing security. No such ingrowth will occur along a length that is completely exposed. Further, an exposed end could cause irritation to soft tissue around the scapula **55**. Further, if the anchor peg **64** were to perforate the bone in an undesirable location the perforation could weaken the scapula and increase the risk of fracture.

[0051] FIG. 1A shows a schematic of a lateral side of a scapula **55**. The glenoid **58** includes an articular surface separated from the rest of the glenoid **58** by a glenoid rim **68**. A healthy shoulder joint will generally have within the glenoid rim **68** an elongate articular surface that has a generally circular inferior portion **70**. More particularly the inferior portion **70** can be bounded by a circular segment of the glenoid rim **68**. The circular portion of the glenoid rim **68** can be disposed about a center **72**. More generally, the center **72** can be a central portion, e.g., a geometric center, of the inferior portion **70**. The center **72** can be disposed on or along an infero-superior axis of the glenoid rim **68** that extends from the superior-most portion (located at the top of the graphic in FIG. 1A) of the glenoid rim **68** to the inferior-most portion (located at the bottom in FIG. 1A) of the glenoid rim **68**. The center **72** may be located at a central portion, e.g., a mid-point, of a chord extending across the glenoid rim **68** at an infero-superior position disposed inferior of a geometric center of the entire glenoid **58** or glenoid rim **68**. For example, the center **72** can be located about one-half to two-thirds of the distance from the inferior-most point of the glenoid rim **68** to the geometric center of the entire glenoid **58** or glenoid rim **68**.

[0052] As will be discussed in greater detail below, an imager **80** can be used to scan the scapula **55** to gather imaging information. That information can be processed in an image processing system **82**. The image processing system **82** can include a memory that can store imaging information corresponding to scanned data from the imager **80**. The image processing system **82** can also include one or more hardware processors that can execute instructions. The image processing system **82** can process the imaging information to identify all the foregoing structures of the scapula **55**. The imaging information can also be processed to locate an anchor trajectory **84** in a direction into the scapula **55** for placement of an anchor peg. The anchor trajectory **84** can be offset from the center **72** of an inferior portion **74** of a glenoid **58**. As shown in FIG. 1A the offset can be in the direction of the anterior surface **76**. The anchor trajectory can be at a center **86** of an opening that can be located between the center **72** and the anterior surface **76**. The center **86** can be located 10% of the distance from the center **72** to the anterior aspect of the glenoid rim **68** adjacent to the anterior surface **76** of the scapula **55**. The center **86** can be located 20% of the distance from the center **72** to the anterior aspect of the glenoid rim **68** adjacent to the anterior surface **76** of the scapula **55**. The center **86** can be located 30% of the distance from the center **72** to the anterior aspect of the glenoid rim **68** adjacent to the anterior surface **76** of the

scapula **55**. The center **86** can be located 40% of the distance from the center **72** to the anterior aspect of the glenoid rim **68** adjacent to the anterior surface **76** of the scapula **55**. The center **86** can be located 50% of the distance from the center **72** to the anterior aspect of the glenoid rim **68** adjacent to the anterior surface **76** of the scapula **55**. The center **86** can be located 60% of the distance from the center **72** to the anterior aspect of the glenoid rim **68** adjacent to the anterior surface **76** of the scapula **55**. In some cases, the center **86** is located in a direction other than toward the anterior surface **76**, e.g., closer to the posterior surface **78** of the scapula **55** by any of these or other percentages. The center **86** could be in other directions as well, e.g., inferior, superior, or some direction between any of anterior, posterior, inferior, or superior depending on the needs of the patient.

[0053] FIG. 1A shows that a patient process flow can enable one to select between a first baseplate **162A** with a centrally positioned anchor peg **164A** and a second baseplate **162B** with an off-set anchor peg **164B**. The first baseplate **162A** is suitable for placement at the center **72**, e.g., when the bone beneath the center **72** has sufficient depth to allow the anchor peg **164A** of the first baseplate **162A** to be enclosed in a blind hole formed in the scapula **55** beneath the center **72**. The second baseplate **162B** is suitable for placement at the center **86**, e.g., when the bone beneath the center **72** does not have sufficient depth but that bone beneath the center **86** has sufficient depth to allow the anchor peg **164B** of the second baseplate **162B** to be enclosed in a blind hole formed in the scapula **55** beneath the center **86**.

[0054] The image processing system **82** can be configured to process imaging information in any suitable manner. FIG. 1B shows one method that can be performed at least in part by the image processing system **82**. In a step **88** a process can receive the imaging information. For example, the imager **80** can be connected by a network to a computer having a processor configured to receive the imaging information. The network can include an Internet connection, a wireless or wired connection within the same facility where the imager **80** is located. In some applications, a data file including the imaging information can be physically transported to an image processing computer. In some applications, the imager **80** is directly connected to a computer with a processor configured to process the imaging information.

[0055] Thereafter in a step **90** a lateral portion or surface **56** of a scapula **55** can be characterized. The characterization of the lateral portion can include segmentation to create a virtual model of all or a portion of the scapula **55**. The step **90** can include forming a virtual model of all or a portion of the humerus **50**. The step **90** can include forming a virtual model of all or a portion of the glenoid **58**. A virtual model formed in step **90** can include a model of the glenoid rim **68**. The virtual model formed in step **90** can include a model of an inferior portion **70** of the glenoid rim **68**. In step **90**, the center **72** of the inferior portion **70** can be identified in the virtual model. The glenoid **58** can be characterized to locate the center **72**, e.g., by obtaining a radius of curvature of the inferior portion **70**. The center **72** can be identified as the center for a radius of curvature of the inferior portion **70**.

[0056] The step **90** can include characterizing a lateral surface **56** of the scapula **55**. The scapula **55** can be disposed in the immediate vicinity of the glenoid **58**, e.g., in lateral facing bone disposed around the glenoid **58**. In some cases, the scapula **55** is further characterized medially of the lateral surface **56**, e.g., along an anterior surface **76** and/or along a

posterior surface **78** of the scapula **55**. The step **90** can include determining the thickness of the scapula **55** between the anterior surface **76** and the posterior surface **78** at one or more locations of the glenoid **58**. For example, thicknesses or depth of bone beneath the glenoid **58** can be determined as a measurement between the surface of the glenoid and an external wall of one of the anterior and posterior surfaces **76**, **78** beneath any point of the glenoid. The depth may be determined relative to the length of a baseplate anchor peg, e.g., less than or greater than such length.

[0057] The step **90** can also include determining a thickness or depth the center **72** and at locations spaced apart from the center **72** if the thickness or depth at the center **72** is not sufficient to fully contain the anchor peg **64** of a glenoid baseplate **62**. The step **90** can identify a range of positions for the placement of an anchor peg **64**, based on more than one position having sufficient scapula bone depth, thickness or quality.

[0058] The image processing system **82** can perform the step **92** in which the location of the center **86** and the anchor trajectory **84** are determined. The image processing system **82** can determine the location of the center **86** by any suitable technique. For example, a hardware processor in the image processing system **82** can execute code implementing a method that determines the thicknesses or depth dimensions for a given location offset from the center **72**. At a location disposed an incremental distance anteriorly from the center **72**, the image processing system **82** can determine the scapula thickness or depth. If the thickness or depth are sufficient for a given patient, the anchor trajectory **84** as well as the location for the center **86** can be established. If the thickness or depth is not sufficient, a further increment from the center **72** can be evaluated by the image processing system **82**. The condition at the bone corresponding to this further increment, e.g., the thickness or depth, can be evaluated by the image processing system **82** to determine if the thickness or depth are sufficient.

[0059] In some embodiments the image processing system **82** performs additional steps of the method of FIG. 1B, e.g., to consider whether to further process the glenoid and/or to generate a configuration for a glenoid baseplate **62**. The configuration of the glenoid baseplate **62** can include an amount of offset between a center of a proximal or distal (or lateral or medial) portion of the glenoid baseplate **62** and the location of the center of the anchor peg **64**. The direction along which the anchor peg **64** extends can be generally perpendicular to a lateral or medial surface of the glenoid baseplate **62** in some embodiments. In some methods, the image processing system **82** concludes the step **92** upon determining the location of the center **86** and the corresponding position of the anchor peg **64** as well as the anchor trajectory **84** within the scapula **55** and the corresponding configuration (e.g., orientation and length) of the anchor peg **64**. The opening **86** may advantageously be determined to be located anterior of the center **72** (e.g., using the first baseplate **162A**) or offset from the center **72** (e.g., using the second baseplate **162B**), e.g., posterior of the center **72**, inferior of the center **72**, superior of the center **72**, or any combination of anterior, posterior, inferior and superior to the center **72** as needed based on the analysis in step **90**. FIGS. 1A shows that when the direction of the center **86** relative to the center **72** is toward the anterior surface **76**, the second baseplate **162B** could be used to maintaining a transverse portion of the baseplate **162B** centered on the

glenoid **58**. A range of positions for the center **86** between the center **72** and the posterior surface **78** also can be provided if the posterior surface **78** extends more generally medially-laterally and the anterior surface **76** is more curved toward the posterior surface **78**.

[0060] In step **93**, the image processing system **82** can determine whether any portion of the glenoid **58** is to be reamed. For example, in some patients, it may be preferable to ream a portion of the glenoid **58** in order to prepare a suitable surface (e.g., a flat or planar surface) for implanting a glenoid baseplate. In some embodiments, the image processing system **82** can comprise processing electronics programmed to automatically determine whether any portion of the glenoid **58** is to be reamed. In other embodiments, the clinician can interact with the image processing system **82** to determine whether any portion of the glenoid **58** is to be reamed. If a determination is made that the glenoid **58** is not to be reamed, then the method moves to block **94**.

[0061] If, however, a determination is made that the glenoid **58** is to be at least partially reamed, then the method moves to a step **95** to determine a location and extent of flat and patient-specific portions of the glenoid **58**. As shown in FIG. 1C, for example, the glenoid **58** can include a patient-specific portion **77** which is to remain unreamed and a reamed portion **75** which is to be reamed before inserting the glenoid baseplate. As shown in FIG. 1C, for example, the reamed portion **75** may comprise only a portion of the entire glenoid **58** surface. The reamed portion **75** can be planarized to facilitate implantation of the glenoid baseplate **62** some patients. In some arrangements, the reamed portion **75** can comprise or surround an osteophyte or other obstructive bone growth or formation. The patient-specific portion **77** can be shaped or contoured to match or substantially match the patient's anatomy that is not to be reamed. Once the location and extent of the reamed and patient specific portions are determined, the method can move to the step **94**.

[0062] In a step **94**, a specification or configuration for a glenoid baseplate **62** (e.g., the first or second baseplates **162A**, **162B**) and for various guides (discussed below in Section I) that can be used to prepare the glenoid **58** and scapula **55** prior to implantation of the baseplate **62** and for glenoid models, various instruments (discussed below in Section I), and other back-table aids (discussed below in Section I) can be output. The output can be in the form of drawings. The output can be computer code to be used by a rapid manufacturing facility. The output in step **94** can be sent directly or indirectly to multiple recipients, including a review recipient, a manufacturing recipient, a physician customer and/or a patient customer.

[0063] In step **96** the configuration or specifications output in step **94** can be received by a manufacturing facility. The configuration or specification can be received by other parties in step **96**. Step **96** can involve a 3D printer of any sort or another form of additive manufacturing receiving instructions output in the step **94**. The instructions can be received and can be implemented by the 3D printer or other additive manufacturing facility forming the glenoid baseplate **62**, guides, instruments, and back-table aids, in a step **98**. In various embodiments, for example, a reaming axis of a reamer and/or reamer guide can be defined at least in part based on the scan and/or 3D model of the patient's scapula. The step **98** generate the glenoid baseplate **62**, guides, instruments, and back-table aids by forming these articles and thereafter putting these articles through appropriate

finishing processes. The step 98 can include transferring the glenoid baseplate 62, guides, instruments, and back-table aides to the surgeon immediately upon concluding the method of FIG. 1B or subsequently.

I. Method of Implanting a Glenoid Baseplate

[0064] Various embodiments disclosed herein relate to methods and instruments for implanting a glenoid baseplate into a scapula of a patient. The methods and instruments can be used to install guide pins in the glenoid 58 using a guide pin placement guide, to partially ream a portion of the glenoid 58, and to implant the glenoid baseplate 62 into the glenoid. A patient-matched anchor peg forming guide can be used to form an anchor channel for the anchor peg of the glenoid baseplate. A screw trajectory guide can be used to form screw holes in the glenoid baseplate.

A. Partial Reaming Guides

[0065] As explained above, in some patients, it may be desirable to only partially ream the glenoid 58. FIG. 2 illustrates a glenoid baseplate 262A having an anchor peg 264 extending therefrom. The glenoid baseplate 262A can have a patient-matched portion 267 having a patient-matched medial side and a planar portion 265 configured to mate with a reamed or otherwise modified glenoid bone surface. The patient-matched portion 267 can be formed using the 3D model of the patient's glenoid 58 as discussed above in connection with FIG. 1B. The planar portion 265 can be substantially flat in order to mate with the partially-reamed surface. For example, as shown in FIG. 3, the glenoid 58 can include the partially reamed portion 75 and the unreamed portion 77. Although the portion 77 will not have been reamed it may be modified in the use of the guides disclosed herein such as by being exposed by removing cartilage. As shown, the reamed portion 75 is only a limited portion of the entire glenoid 58. Thus, when implanted, the planar portion 265 can be positioned against the reamed portion 75, and the patient-specific portion 267 can be positioned against the unreamed natural portion 77 of the glenoid 58.

[0066] To enable partial reaming of the glenoid 58 shown in FIG. 3, a partial reaming guide 300 can be used to ream only a portion of the patient's glenoid 58. As shown in FIGS. 3-3C, the partial reaming guide 300 can include a guide body 301 including a patient-matched surface 302 shaped to conform to an unreamed portion of the scapula of the patient, such as the unreamed portion 77 of the glenoid 58. The partial reaming guide 300 can comprise a reamer depth stop surface 303 at a first height above the patient-matched surface 302. The depth stop surface 303 is configured to be located at a height that sets the depth at which the reaming device planarizes the reamed portion 75 of the scapula. Thus, the reaming device can have a reaming surface that extends below the depth stop surface 303 and can planarize or ream the scapula to form the reamed portion 75. The reamed portion 75 of the scapula can be below the unreamed portion 77 of the scapula following reaming, as shown in FIGS. 2 and 3. Thus, the reamer depth stop surface 303 can be positioned to serve as a depth stop for the reaming device to control a depth and extend of reaming to ensure that only a portion (e.g., reamed portion 75) is reamed from the glenoid 58. Following the use of the guide 300, the glenoid

58 has the combination of reamed 75 and unreamed surface 77 shown in FIG. 3, as discussed further below.

[0067] As shown in FIGS. 3A-3C, a first through hole 304 can be provided in the guide body 301 extending from the reamer depth stop surface 303 to the patient-matched mounting surface 302. The first hole 304 can be aligned with or define a reaming axis 307 about which the reaming device can rotate so as to partially ream the glenoid 58. A receiver body 306, which can comprise a portion of the guide body 301, can extend upwardly from the top surface 303 which also can form a portion of the guide body 301. The receiver body 306 can include a raised surface 309 at a second height. A second hole 305 can extend through the receiver body 306 portion of the guide body 301 from the raised surface 309 to the patient-matched mounting surface 302. The second hole 305 can be offset from the first hole 307 and can be positioned to rotationally orient or align the guide body 301 relative to the scapula of the patient. The second hole 305 can define a rotational alignment axis 308 about which the partial reaming guide 300 can be oriented. The guide body 301 can be advanced over a first guide pin (e.g., guide pin 403 of FIG. 4) through the first hole 304 and a second guide pin (e.g., guide pin 406 of FIG. 4) through the second hole 305. These pins can be aligned or angled to one another in various embodiments. The first and second guide pins can be provided in a manner similar to that explained below in Section I.B.

[0068] Moreover, as shown in FIG. 3C, the first hole 304 can comprise a flared or tapered distal surface 311 that flares outwardly towards the patient-matched mounting surface 302. The second hole 305 can comprise a flared or tapered distal surface 312 that flares outwardly towards the patient-matched mounting surface 302. The tapered surfaces 311, 312 can assist in guiding the partial reaming guide 300 over the first and second guide pins by facilitating insertion of the guide pins in the holes 304, 305.

B. Guide Pin Placement Guides

[0069] It can be important to ensure that the instruments used for preparing the glenoid 58 and implanting the glenoid baseplate 62 are accurately aligned with the appropriate portions of the scapula so as to ensure proper implantation of the baseplate 62. FIG. 4 is a schematic view of a glenoid during a method of preparation of the glenoid 58, in which a guide pin guide 400 has been placed against a rim 68 of the glenoid 58 and/or portions of the scapula around the glenoid 58. The guide pin guide 400 can include a central portion 405 and a plurality of contact members, which can be formed as projections or feet 402 extending from the central portion 405. The feet 402 can be patient-matched such that the feet 402 are positioned against the rim 68 of the glenoid 58. Positioning the feet 402 against the rim 68 of the glenoid 58 can help accurately align the guide 400 with the anatomy. [0070] A first opening 401 can be formed through the central portion 405. As shown in FIG. 4, a first guide pin 403 can be placed through the first opening 401, which can serve as a guide pin hole. The first guide pin 403 can be used to guide instruments over the glenoid 58 during preparation of the scapula for implantation with the baseplate. In various embodiments, the first guide pin 403 can be generally aligned with the reaming axis. One or a plurality of second openings 404 (e.g., 404S, 404A) can be disposed through the guide 400. The second opening(s) 404 can be used to rotationally orient the guide 400 and subsequent instruments

relative to the glenoid **58** and/or to rotationally secure or otherwise affix the guide **400** to the glenoid during placement of the first guide pin **403**. One or more additional guide pins **406**, **406a** can be inserted into one of the plurality of second openings **404S**, **404A**, respectively, at locations offset to one another along the scapula to serve as rotational alignment guides for guiding instruments to the scapula. For example, a superiorly located opening **404S** immediately adjacent to a contact member **402P** disposed on the posterior side of the glenoid **58** can receive the guide pin **406** in one technique. For example, the opening **404S** can also be used for placement of the pin **406** as discussed further below in connection with FIGS. **5** and **6**. In one embodiment of the guide, the opening **404A** can be used for placement of another pin, e.g., the pin **406a** as discussed in connection with FIG. **6**. Once the guide pin **403** (and/or additional guide pins inserted in the second openings **404**) is installed, the guide **400** can be removed from the scapula. The guide pin **403** can remain inserted into the glenoid **58** in order to guide additional instruments to the scapula. The additional guide pins (for example, pins **406** and/or **406a**) can also remain inserted into the glenoid **58** to provide rotational alignment for various treatment instruments.

C. Reamer for Partial Reaming

[0071] As explained above, for some patients, it can be desirable to only partially ream the glenoid **58**. In FIG. **5**, the partial reaming guide **300** can be placed against the surface of the glenoid **58** with the patient-matched surface **302** (see FIGS. **3A-3C**) placed against the glenoid **58**. The first hole **304** of the partial reaming guide **300** can be guided over the guide pin **403** and placed against the glenoid **58**. In some embodiments, the second hole **305** can be guided along a second guide pin **406**. Or, the second guide pin **406** can be placed through the second hole **305** after the guide **300** is placed in contact with the glenoid **58**. The guide pin **403** can be disposed along or parallel to a reaming axis **R** (see FIG. **5A**). The second guide pin **406** can be positioned to rotationally align the partial reaming guide **300** at a desired orientation. In some applications, the desired position for the second guide pin **406** is the same as the location of the (or one of the) second openings **404** in the guide **400**.

[0072] A reaming device **500** (e.g., a pie reamer) can be guided along the guide pin **403**. For example, the reaming device **500** can comprise a stop surface **505** and a reaming portion **502** that includes one or more reaming features, such as blades **503** (FIG. **5A**). An opening **501** can be provided through the reaming device **500**. The opening **501** of the reaming device **500** can be advanced over the guide pin **403** and placed against the scapula. The reaming portion **502** and blades **503** can be placed against the scapula. The reaming device **500** can be rotated about the reaming axis **R**. To enable partial reaming, the reaming device **500** can be reciprocated about the reaming axis **R** about an angle of less than 180 degrees relative to the reaming axis **R**. In some cases, the reaming device **500** can be reciprocated about the reaming axis **R** about an angle of less than 120 degrees relative to the reaming axis **R**. In some cases, the reaming device **500** can be reciprocated about the reaming axis **R** about an angle of less than 90 degrees relative to the reaming axis **R**. In some cases, the reaming device **500** can be reciprocated about the reaming axis **R** about an angle of less than 75 degrees relative to the reaming axis **R**. In some cases, the reaming device **500** can be reciprocated about the

reaming axis **R** about an angle of less than 60 degrees relative to the reaming axis **R**. In some cases, the reaming device **500** can be reciprocated about the reaming axis **R** about an angle of less than 45 degrees relative to the reaming axis **R**. In some cases, the reaming device **500** can be reciprocated about the reaming axis **R** about an angle of less than 30 degrees relative to the reaming axis **R**. In some cases, the reaming device **500** can be reciprocated about the reaming axis **R** about an angle of less than 15 degrees relative to the reaming axis **R**. The reaming device **500** can be reciprocated and advanced until the stop surface **505** of the reaming device **500** contacts the reamer depth stop surface **303** of the partial reaming guide **300**. The reamer depth stop surface **303** of the partial reaming guide **300** and the stop surface **505** of the reaming device **500** can cooperate to ensure that an appropriate depth and extend of the scapula is reamed, based on the patient-specific model of the patient's anatomy. Accordingly, in some applications the reaming device **500** and the reaming guide **300** can be provided to a surgeon together in a kit.

[0073] Because the reaming device **500** may be used in combination with the reaming guide **300** it may be advantageous to provide a narrow profile from one side of the device **500** to an opposite side thereof, e.g., from one end of the blades **503** to another end of the blades. More particularly, the blades **503** can be oriented along an arc **A** that delimits an arc angle and is disposed from a first side **510** of a lower first portion **511** of a body **512** of the device **500** to a second side **513** thereof. The lower first portion **511** can be oriented transverse to an upwardly extending second portion **514** of the body **512**. An angle of the arc **A** between the first side **510** and the second side **513** can be small, e.g., between 10 and 90 degrees, e.g., about 20 degrees, about 30 degrees, about 40 degrees, about 50 degrees, etc.

[0074] The upwardly extending second portion **513** of the body **512** can act as a handle for rotating the reaming device **500** about the reaming axis **R**. In some cases, the reaming device **500** is coupled with or can be part of a driver that can be engaged to oscillate the reaming device **500** by action of a motor or other mechanism.

D. Examples of Reaming Procedures

[0075] FIGS. **5B-5D** illustrate an example method of reaming a portion of a scapula **55**, according to various embodiments. Unless otherwise noted, the components of FIGS. **5B-5D** may be the same as or generally similar to like-numbered components and features of FIGS. **1-5A**, with reference numerals being appended by the letter "A." In the embodiment of FIGS. **5B-5D**, the clinician can ream the glenoid **58** without using a reaming guide. Rather, the clinician can use image data of the patient's glenoid **58** and/or a model of the patient's glenoid **58** to selectively and accurately ream the glenoid **58** to form a reamed portion **75A**, for example, to remove osteophytes and/or asymmetrical bone structure. Omitting the separate reaming guide can reduce the number of components in the kit and accordingly reduce costs.

[0076] In FIG. **5B**, the clinician can use a guide pin guide **400A**, which may be generally similar to or the same as guide pin guide **400**, to ensure accurate placement of a guide pin **403A** (see FIG. **5C**). As explained above, the guide pin guide **400A** can have a bone-facing surface that is patient-matched to the patient's specific, native bone structure, for example, based on image data. The guide pin guide **400A**

can accordingly be placed so as to conform to the contours of the patient's native scapula 55. As above, feet 402A extending from the central portion 405A can be placed against the glenoid 58, and the guide pin 403A can be inserted into the first opening 401A and into the bone.

[0077] Turning to FIG. 5C, the clinician can utilize a reaming device 500A connected to a reamer shaft 525 by way of a tip connector 526. The reaming device 500A can comprise any suitable reaming device, e.g., a standard full reamer, a partial reamer, etc. Instead of using a reaming guide to accurately define a reaming depth, patient-specific image data can be used to select a tip connector 526 that serves to define the depth of the reaming. Examples of the reaming device 500A and tip connector 526 may be similar to those set forth in FIG. 5I below. For example, as explained below, the tip connector 526 may comprise a protrusion 527C that, when pressed against the glenoid 58, defines the depth by which the glenoid 58 is to be reamed, based on the patient-specific image data. For example, based on the image data, the clinician can select a protrusion 527C having a length that, when pressed against the glenoid 58, ensures that the glenoid 58 is reamed to the appropriate depth and at the desired location on the glenoid 58. In some embodiments, the treatment kit can comprise a plurality of tip connectors 526 having different length protrusions 527C, and the clinician can select the appropriate tip connector 526 with a protrusion 527C having a length that provides the appropriate reaming depth based on the patient image data. In other embodiments, the treatment kit can comprise a patient-specific tip connector 526 (e.g., a single tip connector 526) that is manufactured to have a predetermined protrusion length based on patient-specific anatomy. The reaming device 500A can be guided over the guide pin 403A. With the selected tip connector 526 serving as a depth stop, the clinician can activate the reaming device 500A to at least partially ream the glenoid 58. The reaming device 500A can comprise any suitable type of reamer, e.g., a full reamer or a partial (e.g., pie-shaped) reamer. In some embodiments, the reaming device 500A may not be patient-specific. In other embodiments, the reaming device 500A can have a cutting surface or cutting member that is patient-specific, e.g., that is shaped to conform to the patient's anatomy.

[0078] In FIG. 5D, the reaming device 500A can be removed to expose the reamed portion 75A of the glenoid 58. As explained herein, the glenoid assembly 60A can be inserted into the glenoid 58. The glenoid assembly 60A may be the same as or generally similar to the glenoid assembly 60 described herein. For example, the anchor peg 64A and one or multiple screws 65A can anchor the glenoid assembly 60 to the glenoid 58 as explained further herein. Beneficially, the use of the selected tip connector 526 based on patient-specific image data can be used to accurately form the reamed portion 75A without requiring a separate reaming guide.

[0079] FIGS. 5E-5H illustrate an example method of reaming a portion of a scapula 55, according to another embodiment. Unless otherwise noted, the steps and components of FIGS. 5E-5H may be the same as or generally similar to like-numbered components of FIGS. 5B-5D, with reference numerals annotated with the letter "B." For example, as with the method of FIGS. 5B-5D, the guide pin guide 500B can be used to accurately place the guide pin 403B in FIG. 5E. The guide pin guide 500B of FIG. 5E can

be patient-matched to the patient's native bone structure as explained above. In FIG. 5F, as with FIG. 5C, the tip connector 526B can be selected based on patient-specific image data in order to ream to the desired depth.

[0080] Turning to FIG. 5G, in some embodiments, the clinician may want to confirm that the reaming operation has accurately reamed the patient's bone structure, so that the glenoid assembly 60B is accurately positioned. In FIG. 5G, the treatment kit can include a second patient-matched guide pin guide 400C that includes a bone-facing surface that matches to the partially-reamed glenoid 58. For example, the surface of the second guide pin guide 400C can include a patient-matched partial reamed surface that matches the reamed portion 75B of the glenoid 58. If the clinician is satisfied that the patient-matched second guide pin guide 400C accurately matches the reamed portion 75B and the patient's native bone structure, then the clinician can proceed to FIG. 5H and install the glenoid assembly 60B. If the reaming is unsatisfactory, the clinician can continue or modify the reaming until the reamed portion 75B matches the patient-specific guide pin guide 400C.

[0081] FIGS. 5I-5K illustrate example reaming instruments, according to various embodiments. The instrument of FIG. 5I can be the same as or generally similar to the instrument shown in FIG. 5F. For example, FIG. 5I illustrates a reamer shaft 525B, a reaming device 500B, and a tip connector 526B that connects the reaming device 500B to the reamer shaft 525B. The tip connector 526B can be selected to provide a desired clearance between the reaming device 500B and the glenoid 58 based on, e.g., patient-specific image data.

[0082] FIG. 5J is a schematic side sectional view of a reamer shaft 525C, a reaming device 500C, and a tip connector 526C that connects the reaming device 500C to the reamer shaft 525C. The reamer shaft 525C, reaming device 500C, and tip connector 526C can be urged over the guide pin 403C. In the embodiment of FIG. 5J, the tip connector 526C can include a projection 527C that extends distal the reaming device 500C, e.g., distal a distal-most surface of the reaming device 500C. As explained herein, in some embodiments, the reaming device 500C can comprise a patient-specific reaming surface. In other embodiments, the reaming device 500C may not be patient-specific. During reaming, the clinician can urge the projection 527C of the tip connector 526C over the guide pin such that the distal end of the projection 527C contacts the glenoid 58. The projection 527C can accordingly serve to elevate the reaming device 500C above the glenoid 58 by a predetermined amount based on the patient-specific image data. Beneficially, unlike other reaming tips, the projection 527C of the tip connector 526C can serve to enhance lateral reaming, as opposed to medial reaming. For example, based on the patient-specific image data, the clinician can select a tip connector 526C based on a length of the projection 527C, e.g., based on how far beyond the reaming device 500C the projection 527C extends. As shown in FIG. 5J, the projection 527C can position the reaming device 500C above the glenoid 58 such that a first portion 58A (e.g., a lateral portion) of the glenoid 58 is reamed and a second portion 58B of the glenoid is spaced from the reaming device 500C and is not reamed.

[0083] In some embodiments, a treatment kit can include a plurality of tip connectors 526C having projections 527C at different lengths, e.g., at 0.5 mm increments in length.

Based on the image data, the clinician can select a desired tip connector **526C** which can provide lateral reaming at a predetermined depth. In other embodiments, a tip connector **526C** can be manufactured to have a desired length of the projection **527C** based on patient-specific data. The reaming device **500C** and tip connector **526C** can have an opening sized to receive the guide pin **403C**. The reaming device **500C** can be advanced over the guide pin **403C** to at least partially ream the glenoid surface.

[0084] Turning to FIG. 5K, another reaming instrument is disclosed. Unless otherwise noted, the components of FIG. 5K may be similar to like-numbered components of FIG. 5J, with reference numerals appended by the letter “D.” Unlike the embodiment of FIG. 5J, in FIG. 5K, the tip connector **526D** can have a two-stage projection including a first projection **527D** and a second projection **529D** that has a diameter smaller than the diameter of the first projection **527D**. In such embodiments, the clinician may not use a guide pin to ream the glenoid **58**. Rather, the clinician can drill a pilot hole in the glenoid **58**, and can insert the smaller-diameter second projection **529D** of the tip connector **526D** into the pilot hole. The pilot hole can serve as a depth stop such that the first projection **527D** can rest upon the glenoid **58** when the second projection **529D** extends into the pilot hole. Thus, in some embodiments, the clinician can select a tip connector **526D** having lengths for the first and/or second projections **527D**, **529D** selected to provide a suitable amount of reaming based on patient specific data. As above, in some embodiments, the treatment kit can comprise a plurality of differently-sized connectors **526D**. In other embodiments, the kit can comprise a patient-specific connector **526D** manufactured based on patient-specific image data.

E. Patient Matched Anchor Peg Forming Guide

[0085] FIGS. 6-6B illustrate an anchor peg channel forming guide **600** configured to help the clinician form an anchor peg channel **605** in the scapula of the patient. In some methods described herein, the anchor peg channel forming guide **600** is used in a method that can follow the steps illustrated in one or more of FIGS. 4 and 5. In other words, the guide **600** can be used to form an anchor peg channel in a partially reamed glenoid or in a glenoid that has not been reamed.

[0086] The anchor peg channel **605** can be positioned and sized to receive the anchor peg **64** of the glenoid baseplate **62**. The anchor peg channel forming guide **600** can comprise a guide body having a patient-matched surface **606** shaped to conform to the scapula of the patient. A lateral surface **602** can be provided opposite the patient-matched surface **606**. The lateral surface **602** can be generally planar in various embodiments. The anchor peg channel forming guide **600** can include a channel **601** disposed through the guide body of the guide **600**. In various embodiments, the channel **601** can be positioned to be offset from the reaming axis R of the scapula. In various embodiments where no reaming is performed, the channel **601** can be positioned to be offset from a central portion such as may be defined by the first opening **401** of the guide **400**.

[0087] A plurality of rotational alignment holes **603a**, **603b** can also be provided through the guide body of the guide **600**. The rotational alignment holes **603a**, **603b** can be positioned to provide accurate rotational alignment of the anchor peg channel forming guide **600** relative to the

scapula. The rotational alignment holes **603a**, **603b** can be used to secure the guide **600** such that it does not rotate in use or otherwise move. For example, pins **406a**, **406a** can be inserted through the rotational alignment holes **603a**, **603b** to align or to immobilize the guide **600** relative to the scapula. In some variations just one of the holes **603a**, **603b** is present if the guide **600** can be sufficiently stabilized or immobilized without a second of the holes **603a**, **603b**. As shown in FIG. 6, the clinician can use a drill **432** to drill through the channel **601** to form an anchor peg channel, which will have the same size as the inner periphery of the channel **601** and, in the case of an off-set peg baseplate (as in the case of the baseplate **162b**), will be off-set from a hole or axis **605** in the scapula of the patient formed by the guide pin guide **400** through the first opening **401** thereof. In some embodiments, the drill **432** can be advanced over a guide pin (such as guide pin **403**) formed through the guide **400** and can be centered on the reaming axis R. In such cases the guide **600** is optional. Once the anchor peg channel is formed in the glenoid through the channel **601**, the guide **600** can be removed. As shown in FIGS. 6-6B, a center of a periphery of the guide body (in the illustrated embodiment aligned with the axis **605**) can be spaced apart or offset from a center of the channel **601**. The amount and direction of the offset can be patient specific. At least one peripheral hole (e.g., holes **603a** or **603b**) can be used to secure the guide body to the scapula.

[0088] In various embodiments, after the guide **600** has been used to form a channel in the scapula through the channel **601**, the guide **600** can be removed from the scapula and an anchor peg **64** of a baseplate **62** (e.g., the peg **164B** of the baseplate **162B**) can be inserted into the anchor peg channel formed by the guide **600**. As shown in FIG. 7, for example, a tool **63** can be used by the clinician to implant the anchor peg **162B** into the anchor peg channel **605** formed in the glenoid **58**. In other embodiments, the baseplate **162A** can be inserted using the tool **63** if a centered peg channel is formed. A second guide pin **69** (which may be the same as the guide pin **406** or **406a**) can be inserted into the scapula and can provide for rotational alignment for the glenoid baseplate **62**. For example, the second guide pin **69** may be inserted through the peripheral holes **404** of the guide **400** described above.

F. Screw Trajectory Guide

[0089] The glenoid baseplate **62** can be secured to the scapula with one or a plurality of screws. To ensure that the screws are aligned properly relative to the baseplate **62** and scapula, a screw trajectory guide **700** can be mated with the glenoid baseplate **62** to enable the clinician to insert the screws into the scapula of the patient. In some methods the screw trajectories or channels formed along such trajectories are made in a patient specific manner. The screw trajectory guide **700** can include a guide body **701** having a first surface **702a** configured, e.g., shaped to mate with the glenoid baseplate **62** and a second surface **702b** opposite the first surface **702b**. A third surface **702c** can be recessed from the second surface **702b** and can be disposed between the first and second surfaces **702a**, **702b**. As shown in FIGS. 8-8D, the guide body **701** can include one or a plurality of protrusions **704** extending from the first surface **702a** of the guide **700**. The protrusions **704** can be sized and shaped to be inserted into corresponding apertures of the glenoid baseplate **62**.

[0090] As shown in FIGS. 8A, 8B, and 8D, channels 705 can be disposed through the protrusions 704 and can extend from the third surface 702c through the protrusion 704 to a distal end 707 of the protrusion 704. The third surfaces 702c can serve as depth control surfaces and can be disposed at an elevation that is prescribed for the patient to control a depth of a peripheral screw hole formed in the scapula through the channels 705. As shown, axes labeled “Axe2” through “Axe5” can be defined through the channels 705. The axes Axe2 to Axe5 may be non-parallel to one another and selected to secure the baseplate to the scapula. A guide channel 703 can be formed through the guide body 701 so as to define axis Axe1 as shown in FIG. 8A. The guide channel 703 can be shaped and positioned to receive a guide wire or guide pin therethrough in one embodiment. The guide channel 703 can be provided over the guide pin to align the guide 700 with the baseplate 62. In some cases, the baseplate has a tapered proximal member (as shown in connection with baseplate 262 in FIG. 9) which can be received in the guide channel 703. The guide channel 703 can be internally tapered such that a secure connection can be provided between the guide and the baseplate 262. A slot 706 can extend from the guide channel 703 to an outer periphery of the guide body 701. The slot 706 can allow the guide 700 to flex to facilitate removal of the guide 700 from the proximal tapered member of the baseplate 262. The slot 706 can be optional if the connection between the guide 700 and the baseplate 262 does not involve mating tapered surfaces.

[0091] The clinician can drill holes in the patient’s scapula through the peripheral channels 705 and through the corresponding apertures of the glenoid baseplate 62, the baseplate 262 or another baseplate disclosed herein. As explained above, the third surfaces 702c can be recessed to a depth to limit the depth of the corresponding peripheral screw holes. For example, the surfaces 702c can come into contact with a widened portion of a drill being advanced through the channels 705 stopping the drill from being inserted farther than intended.

[0092] Once the holes are drilled, the clinician remove the screw trajectory guide 700. Removing the guide 700 can include flexing the body of the guide about or opposite the slot 706. The clinician can secure the glenoid baseplate 62 to the glenoid 58 using one or a plurality of screws to be inserted through the apertures of the baseplate 62 and the screw holes formed in the scapula through the guide 700 and baseplate.

II. Kits and Systems for Glenoid Preparation and Baseplate Placement

[0093] The components described herein for preparation of the glenoid 58 and implantation of the glenoid baseplate 62 can be incorporated into a system 800 including a kit 800 of at least two of the components described herein. The kit 800 can include any two of the articles shown in FIG. 9 or combinations of these components with other components described herein in other kits.

[0094] For example, as shown in FIG. 9, the kit 800 can include at least two of, e.g., all of the glenoid baseplate 62 (such as any one or more of glenoid baseplates 162A, 162B, 262), the screw trajectory guide 700, the anchor peg forming guide 600, the guide pin guide 400, the partial reaming guide 300, and a three-dimensional (3D) model 900. The 3D model 900 of the patient’s scapula can be based on the scan

of the patient’s anatomy to assist the clinician in preparing the shoulder for surgery. Beneficially, the collection of components in the kit 800 can enable the clinician to prepare and install the glenoid baseplate 62 with all of the components located in one tray or kit. Further, although not shown, the kit 800 can include any other suitable items, such as the reaming device 500 and other system components. The kit 800 can include the partial reaming guide 300 and the reaming device 500. The kit 800 can include the baseplate 162B and the anchor peg channel forming guide 600. The kit 800 can include the baseplate 162A and the other articles shown in FIG. 9 but can exclude the anchor peg channel forming guide 600. Any suitable number of components can be provided in the kit.

Terminology

[0095] As used herein, the relative terms “lateral” and “medial” shall be defined relative to the anatomy. Thus, medial refers to the direction toward the midline and lateral refers to the direction away from the midline.

[0096] Although certain embodiments and examples have been described herein, it will be understood by those skilled in the art that many aspects of the delivery systems shown and described in the present disclosure may be differently combined and/or modified to form still further embodiments or acceptable examples. All such modifications and variations are intended to be included herein within the scope of this disclosure. A wide variety of designs and approaches are possible. No feature, structure, or step disclosed herein is essential or indispensable.

[0097] For purposes of this disclosure, certain aspects, advantages, and novel features are described herein. It is to be understood that not necessarily all such advantages may be achieved in accordance with any particular embodiment. Thus, for example, those skilled in the art will recognize that the disclosure may be embodied or carried out in a manner that achieves one advantage or a group of advantages as taught herein without necessarily achieving other advantages as may be taught or suggested herein.

[0098] Moreover, while illustrative embodiments have been described herein, the scope of any and all embodiments having equivalent elements, modifications, omissions, combinations (e.g., of aspects across various embodiments), adaptations and/or alterations as would be appreciated by those in the art based on the present disclosure. The limitations in the claims are to be interpreted broadly based on the language employed in the claims and not limited to the examples described in the present specification or during the prosecution of the application, which examples are to be construed as non-exclusive. Further, the actions of the disclosed processes and methods may be modified in any manner, including by reordering actions and/or inserting additional actions and/or deleting actions. It is intended, therefore, that the specification and examples be considered as illustrative only, with a true scope and spirit being indicated by the claims and their full scope of equivalents.

[0099] Conditional language used herein, such as, among others, “can,” “might,” “may,” “e.g.,” and the like, unless specifically stated otherwise, or otherwise understood within the context as used, is generally intended to convey that some embodiments include, while other embodiments do not include, certain features, elements, and/or states. Thus, such conditional language is not generally intended to imply that features, elements, blocks, and/or states are in any way

required for one or more embodiments or that one or more embodiments necessarily include logic for deciding, with or without author input or prompting, whether these features, elements and/or states are included or are to be performed in any particular embodiment.

[0100] The ranges disclosed herein also encompass any and all overlap, sub-ranges, and combinations thereof. Language such as “up to,” “at least,” “greater than,” “less than,” “between,” and the like includes the number recited. Numbers preceded by a term such as “about” or “approximately” include the recited numbers and should be interpreted based on the circumstances (e.g., as accurate as reasonably possible under the circumstances, for example $\pm 1\%$, $\pm 5\%$, $\pm 10\%$, $\pm 15\%$, etc.). For example, “about 0.01 inches” includes “0.01 inches.” Phrases preceded by a term such as “substantially” include the recited phrase and should be interpreted based on the circumstances (e.g., as much as reasonably possible under the circumstances). For example, “substantially linear” includes “linear.”

We claim:

1. A surgical guide, comprising:
 - a guide body comprising a patient-matched surface shaped to conform to a portion of a scapula of a patient;
 - a depth stop surface at a first height above the patient-matched surface, the depth stop surface positioned to serve as a depth stop for a reaming device to control a depth of reaming; and
 - wherein the guide body defines a first hole extending from the depth stop surface to the patient-matched surface, the first hole aligned with a reaming axis.
2. The surgical guide of claim 1, wherein the guide body comprises a raised surface at a second height greater than the first height, and wherein the guide body defines a second hole extending from the raised surface to the patient-matched surface.
3. The surgical guide of claim 2, wherein the guide body comprises a receiver body defined between the raised surface and the patient-matched surface.
4. The surgical guide of claim 2, wherein first hole includes a tapered portion that flares outwardly adjacent to the patient-matched surface.
5. The surgical guide of claim 1, wherein the guide body defines a second hole offset from the first hole, the second hole configured to rotationally orient the guide body relative to the scapula.
6. The surgical guide of claim 5, wherein the second hole includes a tapered portion that flares outwardly adjacent to the patient-matched surface.
7. A kit for shoulder surgery, the kit comprising:
 - a first guide configured to limit glenoid reaming about a reaming axis to only a portion of a glenoid, the first guide comprising:
 - a guide body including a patient-matched surface shaped to conform to a portion of a scapula of a patient, and
 - a depth stop surface at a first height above the patient-matched surface, the depth stop surface positioned to serve as a depth stop for a reaming device to control a depth of reaming,

wherein the guide body defines a first hole through extending from the reamer depth stop surface to the patient-matched surface, the first hole aligned with a reaming axis; and

- a second guide having a patient-matched surface shaped to conform to a portion of the glenoid, the second guide having a channel offset from the reaming axis.
8. The kit of claim 7, further comprising:
 - a third guide having a protrusion shaped to be inserted into an aperture of a glenoid baseplate, the protrusion including a channel extending therethrough.
9. The kit of claim 7, further comprising:
 - a glenoid baseplate having an anchor member.
10. The kit of claim 7, further comprising:
 - a three-dimensional (3D) model of the scapula of the patient.
11. The kit of claim 7, further comprising:
 - a third guide having a plurality of contact members configured to conform to a plurality of surfaces of a scapula, the third guide defining an aperture configured for placing a guide wire in the glenoid.
12. The kit of claim 7, wherein the second guide has a portion configured to rest on the reamed portion of the glenoid.
13. The kit of claim 7, further comprising:
 - a reaming device having a body comprising a stop surface and a reaming portion having one or more reaming features configured to ream a bone surface.
14. The kit of claim 13, wherein the body of the reaming device defines a guide hole sized and shaped to receive a guide pin therethrough, and wherein the reaming portion comprises an arc-shaped structure that delimits an angle less than 180 degrees.
15. The kit of claim 14, wherein the reaming device comprises a first lower portion which comprises the reaming portion and a second upper portion angled relative to the lower portion, the guide hole formed through the second upper portion.
16. The kit of claim 7, wherein the guide body of the first guide includes a raised surface at a second height greater than the first height, and wherein the guide body defines a second hole extending from the raised surface to the patient-matched surface.
17. The kit of claim 16, wherein the guide body comprises a receiver body defined between the raised surface and the patient-matched surface.
18. The kit of claim 17, wherein first hole includes a tapered portion that flares outwardly adjacent to the patient-matched surface.
19. The kit of claim 17, wherein the guide body defines a second hole offset from the first hole, the second hole to rotationally orient the guide body relative to the scapula.
20. The kit of claim 19, wherein the second hole includes a tapered portion that flares outwardly adjacent to the patient-matched surface.

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